**EXAMPLE 6** | Use a graphing device to find the values of $x$ for which $e^x > 1{,}000{,}000$.

**SOLUTION** In Figure 19 we graph both the function $y = e^x$ and the horizontal line $y = 1{,}000{,}000$. We see that these curves intersect when $x \approx 13.8$. Thus $e^x > 10^6$ when $x > 13.8$. It is perhaps surprising that the values of the exponential function have already surpassed a million when $x$ is only 14. ■

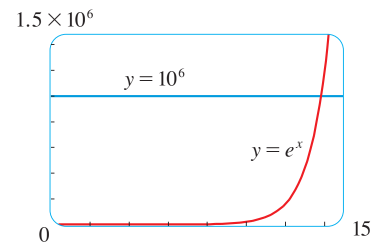



**FIGURE 19**

## EXERCISES 1.4

**1–4** Use the Law of Exponents to rewrite and simplify the expression.

**1.** (a) $\dfrac{4^{-3}}{2^{-8}}$ (b) $\dfrac{1}{\sqrt[3]{x^4}}$

**2.** (a) $8^{4/3}$ (b) $x(3x^2)^3$

**3.** (a) $b^8(2b)^4$ (b) $\dfrac{(6y^3)^4}{2y^5}$

**4.** (a) $\dfrac{x^{2n} \cdot x^{3n-1}}{x^{n+2}}$ (b) $\dfrac{\sqrt{a\sqrt{b}}}{\sqrt[3]{ab}}$

**5.** (a) Write an equation that defines the exponential function with base $b > 0$.
(b) What is the domain of this function?
(c) If $b \neq 1$, what is the range of this function?
(d) Sketch the general shape of the graph of the exponential function for each of the following cases.
(i) $b > 1$ (ii) $b = 1$ (iii) $0 < b < 1$

**6.** (a) How is the number $e$ defined?
(b) What is an approximate value for $e$?
(c) What is the natural exponential function?

**7–10** Graph the given functions on a common screen. How are these graphs related?

**7.** $y = 2^x$, $y = e^x$, $y = 5^x$, $y = 20^x$

**8.** $y = e^x$, $y = e^{-x}$, $y = 8^x$, $y = 8^{-x}$

**9.** $y = 3^x$, $y = 10^x$, $y = \left(\frac{1}{3}\right)^x$, $y = \left(\frac{1}{10}\right)^x$

**10.** $y = 0.9^x$, $y = 0.6^x$, $y = 0.3^x$, $y = 0.1^x$

**11–16** Make a rough sketch of the graph of the function. Do not use a calculator. Just use the graphs given in Figures 4 and 17 and, if necessary, the transformations of Section 1.3.

**11.** $y = 10^{x+2}$

**12.** $y = (0.5)^x - 2$

**13.** $y = -2^{-x}$

**14.** $y = e^{|x|}$

**15.** $y = 1 - \frac{1}{2}e^{-x}$

**16.** $y = 2(1 - e^x)$

**17.** Starting with the graph of $y = e^x$, write the equation of the graph that results from
(a) shifting 2 units downward
(b) shifting 2 units to the right
(c) reflecting about the $x$-axis
(d) reflecting about the $y$-axis
(e) reflecting about the $x$-axis and then about the $y$-axis

**18.** Starting with the graph of $y = e^x$, find the equation of the graph that results from
(a) reflecting about the line $y = 4$
(b) reflecting about the line $x = 2$

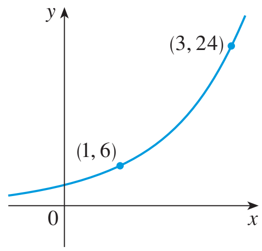
**19–20** Find the domain of each function.

**19.** (a) $f(x) = \dfrac{1 - e^{x^2}}{1 - e^{1-x^2}}$ (b) $f(x) = \dfrac{1 + x}{e^{\cos x}}$

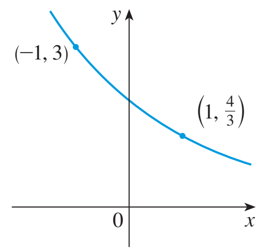
**20.** (a) $g(t) = \sin(e^{-t})$ (b) $g(t) = \sqrt{1 - 2^t}$

**21–22** Find the exponential function $f(x) = Cb^x$ whose graph is given.

**21.**




**22.**




**23.** If $f(x) = 5^x$, show that

$$\frac{f(x+h) - f(x)}{h} = 5^x\left(\frac{5^h - 1}{h}\right)$$

**24.** Suppose you are offered a job that lasts one month. Which of the following methods of payment do you prefer?
  I. One million dollars at the end of the month.
  II. One cent on the first day of the month, two cents on the second day, four cents on the third day, and, in general, $2^{n-1}$ cents on the $n$th day.

**25.** Suppose the graphs of $f(x) = x^2$ and $g(x) = 2^x$ are drawn on a coordinate grid where the unit of measurement is 1 inch. Show that, at a distance 2 ft to the right of the origin, the height of the graph of $f$ is 48 ft but the height of the graph of $g$ is about 265 mi.

**26.** Compare the functions $f(x) = x^5$ and $g(x) = 5^x$ by graphing both functions in several viewing rectangles. Find all points of intersection of the graphs correct to one decimal place. Which function grows more rapidly when $x$ is large?

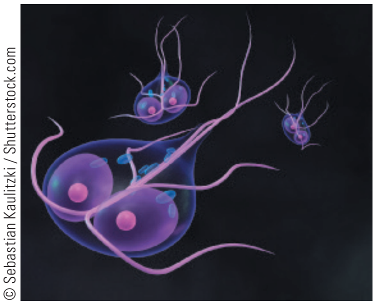
**27.** Compare the functions $f(x) = x^{10}$ and $g(x) = e^x$ by graphing both $f$ and $g$ in several viewing rectangles. When does the graph of $g$ finally surpass the graph of $f$?

**28.** Use a graph to estimate the values of $x$ such that $e^x > 1{,}000{,}000{,}000$.

**29.** ***Giardia lamblia* growth** A researcher is trying to determine the doubling time of a population of the bacterium *Giardia lamblia*. He starts a culture in a nutrient solution and estimates the bacteria count every four hours. His data are shown in the table.

| Time (hours) | 0 | 4 | 8 | 12 | 16 | 20 | 24 |
|---|---|---|---|---|---|---|---|
| Bacteria count (CFU/mL) | 37 | 47 | 63 | 78 | 105 | 130 | 173 |

(a) Make a scatter plot of the data.
(b) Use a graphing calculator to find an exponential curve $f(t) = a \cdot b^t$ that models the bacteria population $t$ hours later.
(c) Graph the model from part (b) together with the scatter plot in part (a). Use the TRACE feature to determine how long it takes for the bacteria count to double.



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*G. lamblia*

**30.** A **bacteria culture** starts with 500 bacteria and doubles in size every half hour.
(a) How many bacteria are there after 3 hours?
(b) How many bacteria are there after $t$ hours?
(c) How many bacteria are there after 40 minutes?
(d) Graph the population function and estimate the time for the population to reach 100,000.

**31.** The half-life of bismuth-210, $^{210}$Bi, is 5 days.
(a) If a sample has a mass of 200 mg, find the amount remaining after 15 days.
(b) Find the amount remaining after $t$ days.
(c) Estimate the amount remaining after 3 weeks.
(d) Use a graph to estimate the time required for the mass to be reduced to 1 mg.

**32.** An isotope of sodium, $^{24}$Na, has a half-life of 15 hours. A sample of this isotope has mass 2 g.
(a) Find the amount remaining after 60 hours.
(b) Find the amount remaining after $t$ hours.
(c) Estimate the amount remaining after 4 days.
(d) Use a graph to estimate the time required for the mass to be reduced to 0.01 g.

**33. Half-life of HIV** Use the graph of $V$ in Figure 12 to estimate the half-life of the viral load of patient 303 during the first month of treatment.

**34. Blood alcohol concentration** After alcohol is fully absorbed into the body, it is metabolized with a half-life of about 1.5 hours. Suppose you have had three alcoholic drinks and an hour later, at midnight, your blood alcohol concentration (BAC) is 0.6 mg/mL.
(a) Find an exponential decay model for your BAC $t$ hours after midnight.
(b) Graph your BAC and use the graph to determine when you can drive home if the legal limit is 0.08 mg/mL.

*Source:* Adapted from P. Wilkinson et al., "Pharmacokinetics of Ethanol after Oral Administration in the Fasting State," *Journal of Pharmacokinetics and Biopharmaceutics* 5 (1977): 207–24.

**35. World population** Use a calculator with exponential regression capability to model the population of the world with the data from 1950 to 2010 in Table 1 on page 46. Use the model to estimate the population in 1993 and to predict the population in the year 2020.

**36. US population** The table gives the population of the United States, in millions, for the years 1900–2010. Use a calculator with exponential regression capability to model the US population since 1900. Use the model to estimate the population in 1925 and to predict the population in the year 2020.

| Year | Population | Year | Population |
|---|---|---|---|
| 1900 | 76 | 1960 | 179 |
| 1910 | 92 | 1970 | 203 |
| 1920 | 106 | 1980 | 227 |
| 1930 | 123 | 1990 | 250 |
| 1940 | 131 | 2000 | 281 |
| 1950 | 150 | 2010 | 310 |

**37.** If you graph the function

$$f(x) = \frac{1 - e^{1/x}}{1 + e^{1/x}}$$

you'll see that $f$ appears to be an odd function. Prove it.

**38.** Graph several members of the family of functions

$$f(x) = \frac{1}{1 + ae^{bx}}$$

where $a > 0$. How does the graph change when $b$ changes? How does it change when $a$ changes?

# 1.5 Logarithms; Semilog and Log-Log Plots

## Inverse Functions

Table 1 gives data from an experiment in which a bacteria culture was started with 100 bacteria in a limited nutrient medium. The size of the bacteria population was recorded at hourly intervals. The number of bacteria $N$ is a function of the time $t$: $N = f(t)$.

Suppose, however, that the biologist changes her point of view and becomes interested in the time required for the population to reach various levels. In other words, she is thinking of $t$ as a function of $N$. This function is called the *inverse function* of $f$, denoted by $f^{-1}$, and read "$f$ inverse." Thus $t = f^{-1}(N)$ is the time required for the population level to reach $N$. The values of $f^{-1}$ can be found by reading Table 1 from right to left or by consulting Table 2. For instance, $f^{-1}(550) = 6$ because $f(6) = 550$.

**Table 1** $N$ as a function of $t$

| $t$ (hours) | $N = f(t)$ = population at time $t$ |
|---|---|
| 0 | 100 |
| 1 | 168 |
| 2 | 259 |
| 3 | 358 |
| 4 | 445 |
| 5 | 509 |
| 6 | 550 |
| 7 | 573 |
| 8 | 586 |

**Table 2** $t$ as a function of $N$

| $N$ | $t = f^{-1}(N)$ = time to reach $N$ bacteria |
|---|---|
| 100 | 0 |
| 168 | 1 |
| 259 | 2 |
| 358 | 3 |
| 445 | 4 |
| 509 | 5 |
| 550 | 6 |
| 573 | 7 |
| 586 | 8 |

Not all functions possess inverses. Let's compare the functions $f$ and $g$ whose arrow diagrams are shown in Figure 1. Note that $f$ never takes on the same value twice (any two inputs in $A$ have different outputs), whereas $g$ does take on the same value twice (both 2 and 3 have the same output, 4). In symbols,

$$g(2) = g(3)$$

but $$f(x_1) \neq f(x_2) \qquad \text{whenever } x_1 \neq x_2$$

Functions that share this property with $f$ are called *one-to-one functions*.

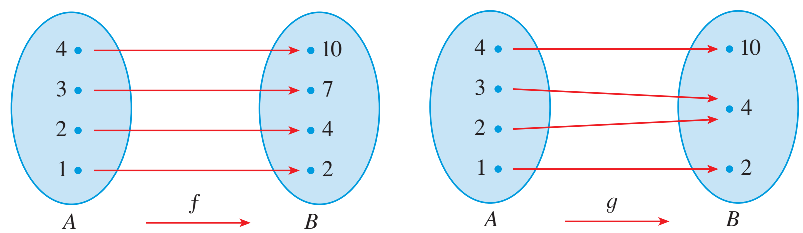



**FIGURE 1**
$f$ is one-to-one; $g$ is not.

In the language of inputs and outputs, this definition says that $f$ is one-to-one if each output corresponds to only one input.

**(1) Definition** A function $f$ is called a **one-to-one function** if it never takes on the same value twice; that is,

$$f(x_1) \neq f(x_2) \qquad \text{whenever } x_1 \neq x_2$$

If a horizontal line intersects the graph of $f$ in more than one point, then we see from Figure 2 that there are numbers $x_1$ and $x_2$ such that $f(x_1) = f(x_2)$. This means that $f$ is not one-to-one. Therefore we have the following geometric method for determining whether a function is one-to-one.

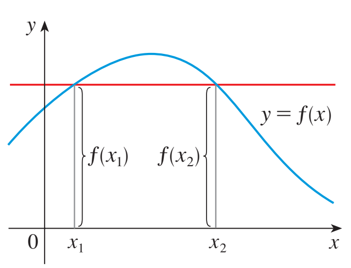



**FIGURE 2**
The function is not one-to-one because $f(x_1) = f(x_2)$.

**Horizontal Line Test** A function is one-to-one if and only if no horizontal line intersects its graph more than once.

**EXAMPLE 1** | Is the function $f(x) = x^3$ one-to-one?

**SOLUTION 1** If $x_1 \neq x_2$, then $x_1^3 \neq x_2^3$ (two different numbers can't have the same cube). Therefore, by Definition 1, $f(x) = x^3$ is one-to-one.

**SOLUTION 2** From Figure 3 we see that no horizontal line intersects the graph of $f(x) = x^3$ more than once. Therefore, by the Horizontal Line Test, $f$ is one-to-one. ■

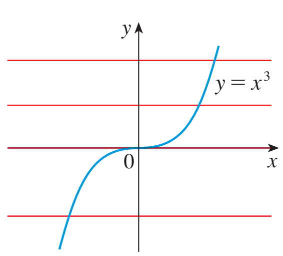



**FIGURE 3**
$f(x) = x^3$ is one-to-one.

**EXAMPLE 2** | Is the function $g(x) = x^2$ one-to-one?

**SOLUTION 1** This function is not one-to-one because, for instance,

$$g(1) = 1 = g(-1)$$

and so 1 and $-1$ have the same output.

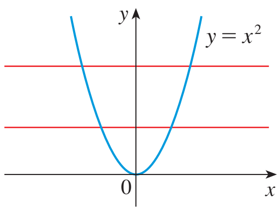



**FIGURE 4**
$g(x) = x^2$ is not one-to-one.

**SOLUTION 2** From Figure 4 we see that there are horizontal lines that intersect the graph of $g$ more than once. Therefore, by the Horizontal Line Test, $g$ is not one-to-one. ■

One-to-one functions are important because they are precisely the functions that possess inverse functions according to the following definition.

> **(2) Definition** Let $f$ be a one-to-one function with domain $A$ and range $B$. Then its **inverse function** $f^{-1}$ has domain $B$ and range $A$ and is defined by
>
> $$f^{-1}(y) = x \iff f(x) = y$$
>
> for any $y$ in $B$.

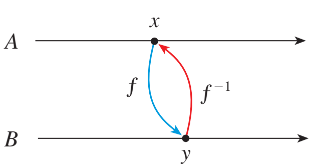



**FIGURE 5**

This definition says that if $f$ maps $x$ into $y$, then $f^{-1}$ maps $y$ back into $x$. (If $f$ were not one-to-one, then $f^{-1}$ would not be uniquely defined.) The arrow diagram in Figure 5 indicates that $f^{-1}$ reverses the effect of $f$. Note that

$$\boxed{\begin{aligned} \text{domain of } f^{-1} &= \text{range of } f \\ \text{range of } f^{-1} &= \text{domain of } f \end{aligned}}$$

For example, the inverse function of $f(x) = x^3$ is $f^{-1}(x) = x^{1/3}$ because if $y = x^3$, then

$$f^{-1}(y) = f^{-1}(x^3) = (x^3)^{1/3} = x$$

**CAUTION** Do not mistake the $-1$ in $f^{-1}$ for an exponent. Thus

$$f^{-1}(x) \quad \text{does } \textit{not} \text{ mean} \quad \frac{1}{f(x)}$$

The reciprocal $1/f(x)$ could, however, be written as $[f(x)]^{-1}$.

**EXAMPLE 3** | If $f$ is one-to-one and $f(1) = 5$, $f(3) = 7$, and $f(8) = -10$, find $f^{-1}(7)$, $f^{-1}(5)$, and $f^{-1}(-10)$.

**SOLUTION** From the definition of $f^{-1}$ we have

$$f^{-1}(7) = 3 \qquad \text{because} \qquad f(3) = 7$$

$$f^{-1}(5) = 1 \qquad \text{because} \qquad f(1) = 5$$

$$f^{-1}(-10) = 8 \qquad \text{because} \qquad f(8) = -10$$

The diagram in Figure 6 makes it clear how $f^{-1}$ reverses the effect of $f$ in this case.

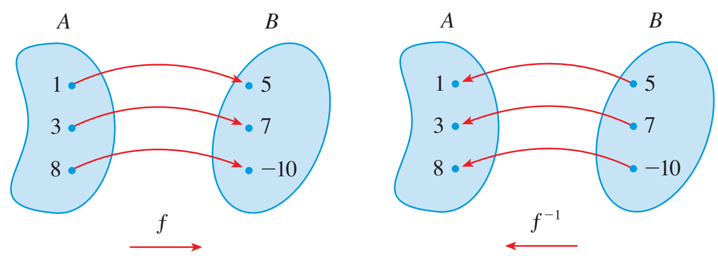



**FIGURE 6**
The inverse function reverses inputs and outputs.

The letter $x$ is traditionally used as the independent variable, so when we concentrate on $f^{-1}$ rather than on $f$, we usually reverse the roles of $x$ and $y$ in Definition 2 and write

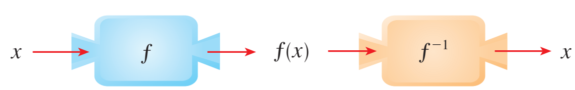
**(3)**
$$\boxed{f^{-1}(x) = y \iff f(y) = x}$$

By substituting for $y$ in Definition 2 and substituting for $x$ in (3), we get the following **cancellation equations**:

**(4)**
$$f^{-1}(f(x)) = x \quad \text{for every } x \text{ in } A$$
$$f(f^{-1}(x)) = x \quad \text{for every } x \text{ in } B$$

The first cancellation equation says that if we start with $x$, apply $f$, and then apply $f^{-1}$, we arrive back at $x$, where we started (see the machine diagram in Figure 7). Thus $f^{-1}$ undoes what $f$ does. The second equation says that $f$ undoes what $f^{-1}$ does.

**FIGURE 7**




For example, if $f(x) = x^3$, then $f^{-1}(x) = x^{1/3}$ and so the cancellation equations become

$$f^{-1}(f(x)) = (x^3)^{1/3} = x$$

$$f(f^{-1}(x)) = (x^{1/3})^3 = x$$

These equations simply say that the cube function and the cube root function cancel each other when applied in succession.

Now let's see how to compute inverse functions. If we have a function $y = f(x)$ and are able to solve this equation for $x$ in terms of $y$, then according to Definition 2 we must have $x = f^{-1}(y)$. If we want to call the independent variable $x$, we then interchange $x$ and $y$ and arrive at the equation $y = f^{-1}(x)$.

**(5) How to Find the Inverse Function of a One-to-One Function $f$**

**STEP 1** Write $y = f(x)$.

**STEP 2** Solve this equation for $x$ in terms of $y$ (if possible).

**STEP 3** To express $f^{-1}$ as a function of $x$, interchange $x$ and $y$.
The resulting equation is $y = f^{-1}(x)$.

**EXAMPLE 4** | Find the inverse function of $f(x) = x^3 + 2$.

**SOLUTION** According to (5) we first write

$$y = x^3 + 2$$

Then we solve this equation for $x$:

$$x^3 = y - 2$$

$$x = \sqrt[3]{y - 2}$$

Finally, we interchange $x$ and $y$:

$$y = \sqrt[3]{x - 2}$$

Therefore the inverse function is $f^{-1}(x) = \sqrt[3]{x - 2}$. ■

In Example 4, notice how $f^{-1}$ reverses the effect of $f$. The function $f$ is the rule "Cube, then add 2"; $f^{-1}$ is the rule "Subtract 2, then take the cube root."

The principle of interchanging $x$ and $y$ to find the inverse function also gives us the method for obtaining the graph of $f^{-1}$ from the graph of $f$. Since $f(a) = b$ if and only if $f^{-1}(b) = a$, the point $(a, b)$ is on the graph of $f$ if and only if the point $(b, a)$ is on the graph of $f^{-1}$. But we get the point $(b, a)$ from $(a, b)$ by reflecting about the line $y = x$. (See Figure 8.)

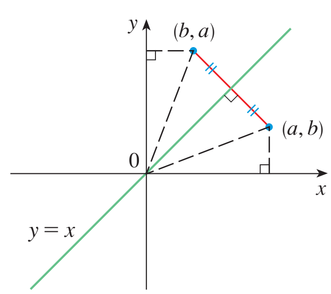



**FIGURE 8**

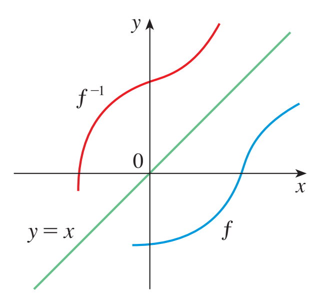



**FIGURE 9**

Therefore, as illustrated by Figure 9:

> The graph of $f^{-1}$ is obtained by reflecting the graph of $f$ about the line $y = x$.

**EXAMPLE 5** | **Reproduction number** One of the main quantities that epidemiologists try to measure for infectious diseases is the so-called *basic reproduction number*, denoted by $R_0$. Biologically, this is the expected number of new infections that an infected individual will produce when introduced into a completely susceptible population. The probability of an outbreak occurring (as opposed to the disease dying out by chance) is then modeled by the equation

$$P = 1 - \frac{1}{R_0} \qquad \text{where } R_0 \geqslant 1$$

Find the inverse function of $P$, interpret it, and graph it.

Ranges of $R_0$ values for some common diseases

| Disease | $R_0$ |
|---|---|
| Measles | 12–18 |
| Diphtheria | 6–7 |
| Smallpox | 5–7 |
| Polio | 5–7 |
| Mumps | 4–7 |
| HIV | 2–5 |

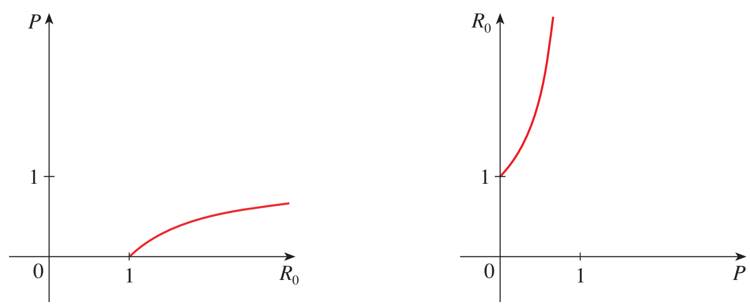
**SOLUTION** To find the inverse function we solve the given equation for $R_0$:

$$\frac{1}{R_0} = 1 - P \qquad\qquad R_0 = \frac{1}{1 - P}$$

This equation expresses $R_0$ as a function of $P$. It gives the value of $R_0$ that is required to obtain outbreak probability $P$.

In Exercise 32 this model is modified to take vaccinations into account.

Notice that we did not interchange the variables $P$ and $R_0$ because they need to retain their meanings. Figure 10 shows the graphs of each of these variables as a function of the other variable. Observe that each of these graphs is a reflection of the other about the diagonal line $P = R_0$.




(a) $P$ as a function of $R_0$

$P = 1 - \dfrac{1}{R_0}, \quad R_0 \geqslant 1$

(b) $R_0$ as a function of $P$

$R_0 = \dfrac{1}{1 - P}, \quad 0 \leqslant P < 1$

**FIGURE 10**

■

## ■ Logarithmic Functions

If $b > 0$ and $b \neq 1$, the exponential function $f(x) = b^x$ is either increasing or decreasing and so it is one-to-one by the Horizontal Line Test. It therefore has an inverse function $f^{-1}$, which is called the **logarithmic function with base $b$** and is denoted by $\log_b$. If we use the formulation of an inverse function given by (3),

$$f^{-1}(x) = y \iff f(y) = x$$

then we have

**(6)**
$$\boxed{\log_b x = y \iff b^y = x}$$

Thus, if $x > 0$, then $\log_b x$ is the exponent to which the base $b$ must be raised to give $x$. For example, $\log_{10} 0.001 = -3$ because $10^{-3} = 0.001$.

The cancellation equations (4), when applied to the functions $f(x) = b^x$ and $f^{-1}(x) = \log_b x$, become

**(7)**
$$\boxed{\begin{aligned} \log_b(b^x) &= x \quad \text{for every } x \in \mathbb{R} \\ b^{\log_b x} &= x \quad \text{for every } x > 0 \end{aligned}}$$

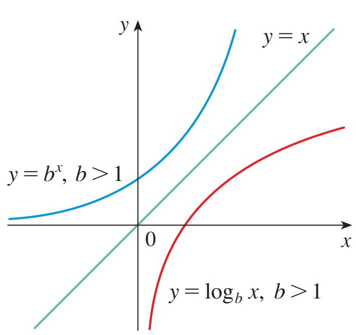



**FIGURE 11**

The logarithmic function $\log_b$ has domain $(0, \infty)$ and range $\mathbb{R}$. Its graph is the reflection of the graph of $y = b^x$ about the line $y = x$.

Figure 11 shows the case where $b > 1$. (The most important logarithmic functions have base $b > 1$.) The fact that $y = b^x$ is a very rapidly increasing function for $x > 0$ is reflected in the fact that $y = \log_b x$ is a very slowly increasing function for $x > 1$.

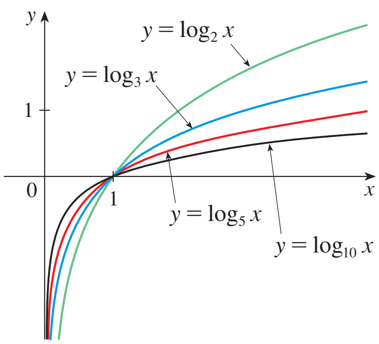



**FIGURE 12**

Figure 12 shows the graphs of $y = \log_b x$ with various values of the base $b > 1$. Since $\log_b 1 = 0$, the graphs of all logarithmic functions pass through the point $(1, 0)$.

The following properties of logarithmic functions follow from the corresponding properties of exponential functions given in Section 1.4.

**Laws of Logarithms** If $x$ and $y$ are positive numbers, then

**1.** $\log_b(xy) = \log_b x + \log_b y$

**2.** $\log_b\left(\dfrac{x}{y}\right) = \log_b x - \log_b y$

**3.** $\log_b(x^r) = r\log_b x$ (where $r$ is any real number)

**EXAMPLE 6** | Use the laws of logarithms to evaluate $\log_2 80 - \log_2 5$.

**SOLUTION** Using Law 2, we have

$$\log_2 80 - \log_2 5 = \log_2\left(\frac{80}{5}\right) = \log_2 16 = 4$$

because $2^4 = 16$. ■

## Natural Logarithms

**Notation for Logarithms**
Most textbooks in calculus and the sciences, as well as calculators, use the notation $\ln x$ for the natural logarithm and $\log x$ for the "common logarithm," $\log_{10} x$. In the more advanced mathematical and scientific literature and in computer languages, however, the notation $\log x$ usually denotes the natural logarithm.

Of all possible bases $b$ for logarithms, we will see in Chapter 3 that the most convenient choice of a base is the number $e$, which was defined in Section 1.4. The logarithm with base $e$ is called the **natural logarithm** and has a special notation:

$$\boxed{\log_e x = \ln x}$$

If we put $b = e$ and replace $\log_e$ with "ln" in (6) and (7), then the defining properties of the natural logarithm function become

**(8)**
$$\boxed{\ln x = y \iff e^y = x}$$

**(9)**
$$\boxed{\begin{aligned} \ln(e^x) &= x \quad \text{for every } x \in \mathbb{R} \\ e^{\ln x} &= x \quad \text{for every } x > 0 \end{aligned}}$$

In particular, if we set $x = 1$, we get

$$\boxed{\ln e = 1}$$

**EXAMPLE 7** | Find $x$ if $\ln x = 5$.

**SOLUTION 1** From (8) we see that

$$\ln x = 5 \qquad \text{means} \qquad e^5 = x$$

Therefore $x = e^5$.

(If you have trouble working with the "ln" notation, just replace it by $\log_e$. Then the equation becomes $\log_e x = 5$; so, by the definition of logarithm, $e^5 = x$.)

**SOLUTION 2** Start with the equation

$$\ln x = 5$$

and apply the exponential function to both sides of the equation:

$$e^{\ln x} = e^5$$

But the second cancellation equation in (9) says that $e^{\ln x} = x$. Therefore $x = e^5$. ■

**EXAMPLE 8** | Solve the equation $e^{5-3x} = 10$.

**SOLUTION** We take natural logarithms of both sides of the equation and use (9):

$$\begin{aligned} \ln(e^{5-3x}) &= \ln 10 \\ 5 - 3x &= \ln 10 \\ 3x &= 5 - \ln 10 \\ x &= \tfrac{1}{3}(5 - \ln 10) \end{aligned}$$

Since the natural logarithm is found on scientific calculators, we can approximate the solution: to four decimal places, $x \approx 0.8991$. ■

**EXAMPLE 9** | Express $\ln a + \frac{1}{2}\ln b$ as a single logarithm.

**SOLUTION** Using Laws 3 and 1 of logarithms, we have

$$\begin{aligned} \ln a + \tfrac{1}{2}\ln b &= \ln a + \ln b^{1/2} \\ &= \ln a + \ln\sqrt{b} \\ &= \ln\left(a\sqrt{b}\right) \end{aligned}$$ ■

The following formula shows that logarithms with any base can be expressed in terms of the natural logarithm.

**(10) Change of Base Formula** For any positive number $b$ ($b \neq 1$), we have

$$\log_b x = \frac{\ln x}{\ln b}$$

**PROOF** Let $y = \log_b x$. Then, from (6), we have $b^y = x$. Taking natural logarithms of both sides of this equation, we get $y \ln b = \ln x$. Therefore

$$y = \frac{\ln x}{\ln b}$$ ■

Scientific calculators have a key for natural logarithms, so Formula 10 enables us to use a calculator to compute a logarithm with any base (as shown in the following example).

**EXAMPLE 10** | Evaluate $\log_8 5$ correct to six decimal places.

**SOLUTION** Formula 10 gives

$$\log_8 5 = \frac{\ln 5}{\ln 8} \approx 0.773976$$

■

**EXAMPLE 11** | In Example 1.4.4 we showed that the mass of $^{90}$Sr that remains from a 24-mg sample after $t$ years is $m = f(t) = 24 \cdot 2^{-t/25}$. Find the inverse of this function and interpret it.

**SOLUTION** We need to solve the equation $m = f(t) = 24 \cdot 2^{-t/25}$ for $t$. We start by isolating the exponential and taking natural logarithms of both sides:

$$2^{-t/25} = \frac{m}{24}$$

$$\ln(2^{-t/25}) = \ln\left(\frac{m}{24}\right)$$

$$-\frac{t}{25}\ln 2 = \ln m - \ln 24$$

$$t = -\frac{25}{\ln 2}(\ln m - \ln 24) = \frac{25}{\ln 2}(\ln 24 - \ln m)$$

So the inverse function is

$$f^{-1}(m) = \frac{25}{\ln 2}(\ln 24 - \ln m)$$

This function gives the time required for the mass to decay to $m$ milligrams. In particular, the time required for the mass to be reduced to 5 mg is

$$t = f^{-1}(5) = \frac{25}{\ln 2}(\ln 24 - \ln 5) \approx 56.58 \text{ years}$$

This answer agrees with the graphical estimate that we made in Example 1.4.4(c). ■

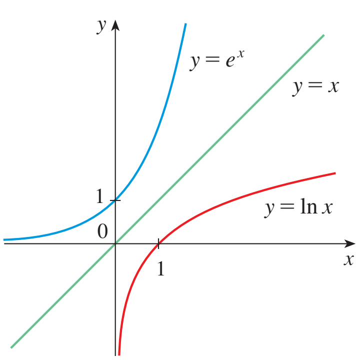



**FIGURE 13**
The graph of $y = \ln x$ is the reflection of the graph of $y = e^x$ about the line $y = x$.

## ■ Graph and Growth of the Natural Logarithm

The graphs of the exponential function $y = e^x$ and its inverse function, the natural logarithm function, are shown in Figure 13. Because the curve $y = e^x$ crosses the $y$-axis with a slope of 1, it follows that the reflected curve $y = \ln x$ crosses the $x$-axis with a slope of 1.

In common with all other logarithmic functions with base greater than 1, the natural logarithm is an increasing function defined on $(0, \infty)$ and the $y$-axis is a vertical asymptote. (This means that the values of $\ln x$ become very large negative as $x$ approaches 0.)

**EXAMPLE 12** | Sketch the graph of the function $y = \ln(x - 2) - 1$.

**SOLUTION** We start with the graph of $y = \ln x$ as given in Figure 13. Using the transformations of Section 1.3, we shift it 2 units to the right to get the graph of $y = \ln(x - 2)$ and then we shift it 1 unit downward to get the graph of $y = \ln(x - 2) - 1$. (See Figure 14.)

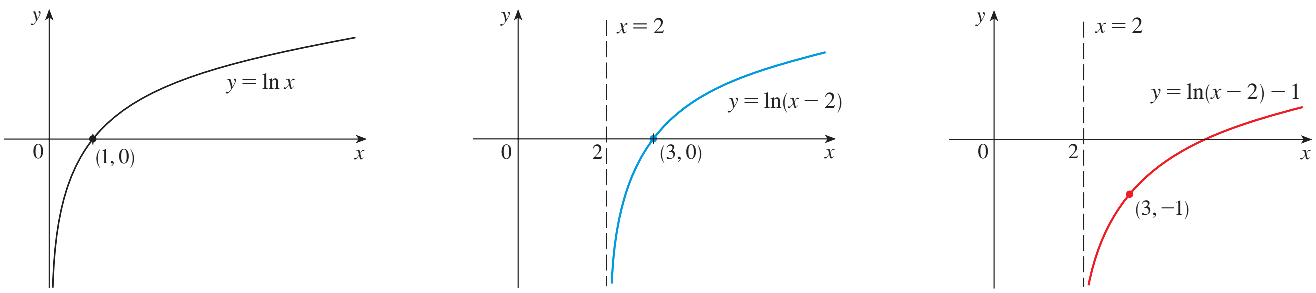



**FIGURE 14**

Although $\ln x$ is an increasing function, it grows *very* slowly when $x > 1$. In fact, $\ln x$ grows more slowly than any positive power of $x$. To illustrate this fact, we compare approximate values of the functions $y = \ln x$ and $y = x^{1/2} = \sqrt{x}$ in the following table and we graph them in Figures 15 and 16. You can see that initially the graphs of $y = \sqrt{x}$ and $y = \ln x$ grow at comparable rates, but eventually the root function far surpasses the logarithm.

| $x$ | 1 | 2 | 5 | 10 | 50 | 100 | 500 | 1000 | 10,000 | 100,000 |
|---|---|---|---|---|---|---|---|---|---|---|
| $\ln x$ | 0 | 0.69 | 1.61 | 2.30 | 3.91 | 4.6 | 6.2 | 6.9 | 9.2 | 11.5 |
| $\sqrt{x}$ | 1 | 1.41 | 2.24 | 3.16 | 7.07 | 10.0 | 22.4 | 31.6 | 100 | 316 |
| $\dfrac{\ln x}{\sqrt{x}}$ | 0 | 0.49 | 0.72 | 0.73 | 0.55 | 0.46 | 0.28 | 0.22 | 0.09 | 0.04 |

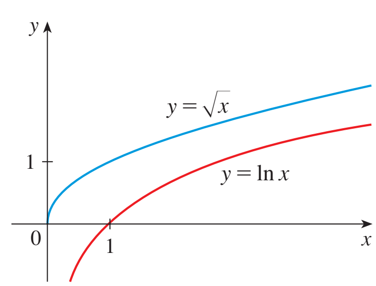



**FIGURE 15**

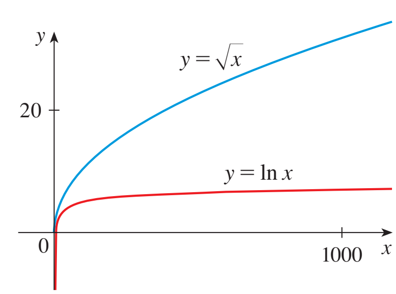



**FIGURE 16**

## Semilog Plots

We've seen that the exponential function $y = b^x$ ($b > 1$) increases so rapidly that it's sometimes difficult to represent data points conveniently on a single plot. (See Figure 1.4.1.) On the other hand, we have just seen that their inverse functions, the logarithmic functions, increase very slowly. For that reason, logarithmic scales are often used when real-world quantities involve a huge disparity in size: the pH scale for the acidity of a solution, the decibel scale for loudness, the Richter scale for the magnitude of an earthquake. In such cases, the equidistant marks on a **logarithmic scale** represent consecutive powers of 10. (See Figure 17.)

The marks on the "logarithmic ruler" are the logarithms of the numbers they represent.

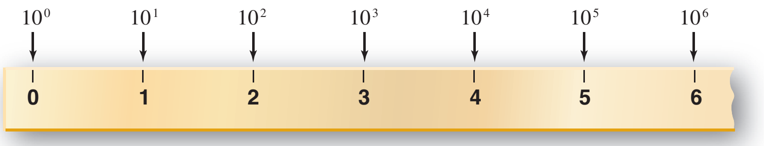



**FIGURE 17**

Vast differences in size occur in biology too. Figure 18 shows that, in comparing lengths, a logarithmic scale provides more manageable numbers.

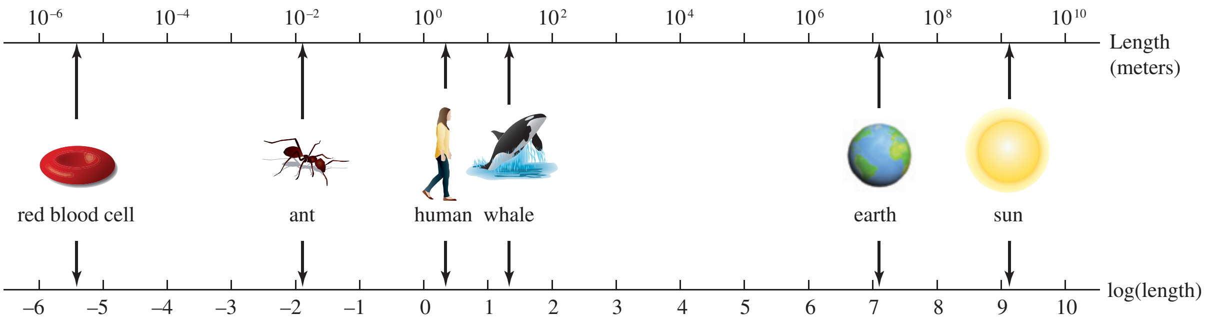



**FIGURE 18**

In biology it's common to use a **semilog plot** to see whether data points are appropriately modeled by an exponential function. This means that instead of plotting the points $(x, y)$, we plot the points $(x, \log y)$. (Logarithms to the base 10 are always used, so $\log = \log_{10}$.) In other words, we use a logarithmic scale on the vertical axis.

If we start with an exponential function of the form $y = a \cdot b^x$ and take logarithms of both sides, we get

$$\log y = \log(a \cdot b^x) = \log a + \log b^x$$

$$\log y = \log a + x \log b \tag{11}$$

If we let $Y = \log y$, $M = \log b$, and $B = \log a$, then Equation 11 becomes

$$Y = B + Mx$$

which is the equation of a line with slope $M$ and $Y$-intercept $B$.

So if we obtain experimental data that we suspect might possibly be exponential, then we could graph a semilog scatter plot and see if it is approximately linear. If so, we could then obtain an exponential model for our original data.

**EXAMPLE 13** | Viral load In Section 1.4 we presented data on the viral load $V(t)$ of patient 303 after $t$ days of treatment with ABT-538. In the following table we calculate $\log V(t)$ and in Figure 19 we show the corresponding semilog plot.

| $t$ | $V(t)$ | $\log V(t)$ |
|---|---|---|
| 1 | 76 | 1.9 |
| 4 | 53 | 1.7 |
| 8 | 18 | 1.3 |
| 11 | 9.4 | 0.97 |
| 15 | 5.2 | 0.72 |
| 22 | 3.6 | 0.56 |
| 29 | 2.8 | 0.45 |

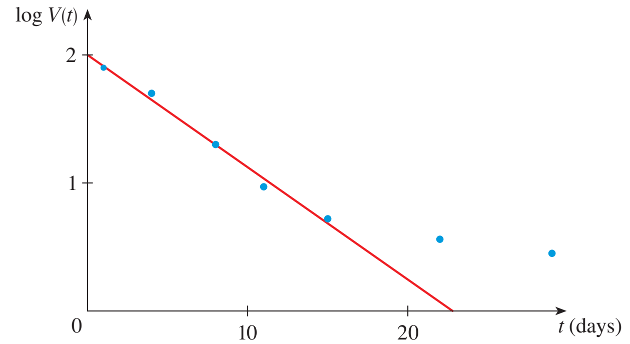



**FIGURE 19**

We see that the first five data points in Figure 19 lie very nearly on a straight line and, using linear regression, we get the equation

$$\log V(t) = 2.006 - 0.088t$$

for $1 \leqslant t \leqslant 15$. Then, applying the exponential function with base 10 to both sides of this equation, we obtain an equation for the viral load:

$$V(t) = 101 \cdot (0.817)^t$$

This is quite close to the exponential model we got in Section 1.4 using six data points. ■

## ■ Log-Log Plots

If we use logarithmic scales on both the horizontal and vertical axes, the resulting graph is called a **log-log plot**. It is used when we suspect that a power function might be a good model for our data. If we start with a power function $y = Cx^p$ and take logarithms of both sides, we get

$$\log y = \log(Cx^p) = \log C + \log x^p$$

$$\log y = \log C + p \log x \tag{12}$$

Let $Y = \log y$, $A = \log C$, and $X = \log x$. Then Equation 12 becomes

$$Y = A + pX$$

We recognize that $Y$ is a linear function of $X$, so the points $(\log x, \log y)$ lie on a straight line.

**EXAMPLE 14** | **BB** Species richness in bat caves Table 3 on page 64 gives the areas of several caves in central Mexico and the number of bat species that live in each cave.[1]

(a) Make a scatter plot and a log-log plot of the data.
(b) Is a power model appropriate? If so, find an expression for it.
(c) The cave called El Sapo near Puebla, Mexico, has a surface area of $A = 205\ \text{m}^2$. Use the model to estimate the number of bat species you would expect to find in that cave.

1. A. Brunet et al., "The Species–Area Relationship in Bat Assemblages of Tropical Caves," *Journal of Mammalogy* 82 (2001): 1114–22.

**Table 3** Species–Area Data

| Cave | Area ($m^2$) | Number of Species |
|---|---|---|
| La Escondida | 18 | 1 |
| El Escorpion | 19 | 1 |
| El Tigre | 58 | 1 |
| Misión Imposible | 60 | 2 |
| San Martin | 128 | 5 |
| El Arenal | 187 | 4 |
| La Ciudad | 344 | 6 |
| Virgen | 511 | 7 |

**SOLUTION**

(a) Let $A$ denote the surface area of a cave and $S$ the number of bat species in the cave. From the scatter plot in Figure 20 we see that the data are neither linear nor exponential. So we calculate the logarithms of the data and create the log-log plot in Figure 21.

| log $A$ | log $S$ |
|---|---|
| 1.26 | 0 |
| 1.28 | 0 |
| 1.76 | 0 |
| 1.78 | 0.30 |
| 2.11 | 0.70 |
| 2.27 | 0.60 |
| 2.54 | 0.78 |
| 2.71 | 0.85 |

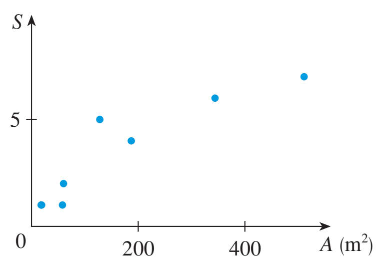



**FIGURE 20**

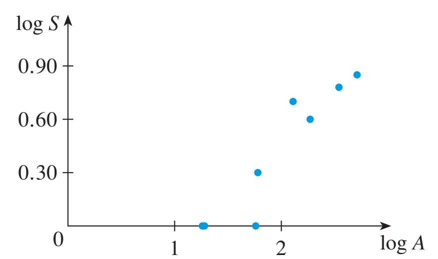



**FIGURE 21**

(b) It appears that log $S$ is approximately a linear function of log $A$. With a graphing calculator or computer, we get the linear model

$$\log S = 0.64 \log A - 0.86$$

Then we apply the exponential function with base 10 to both sides of this equation:

$$S = 10^{0.64 \log A - 0.86} = 10^{\log A^{0.64}} 10^{-0.86}$$
$$S = 0.14A^{0.64}$$

(Alternatively, after verifying from Figure 21 that a power model is appropriate, we could have used a calculator to calculate this power model directly from the original data.) Figure 22 shows that the power model is a reasonable one.

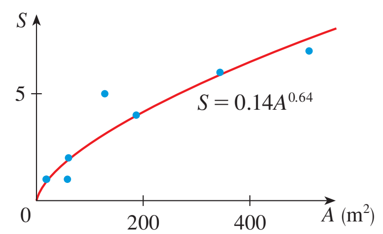



**FIGURE 22**

(c) Using the model from part (b) with $A = 205$, we get

$$S \approx 0.14 \cdot 205^{0.64} \approx 4.22$$

So we would expect to find about four bat species in the El Sapo cave. ■

The El Sapo cave actually does have four species of bats.

**Summary: Linear, Exponential, or Power Model?**

To determine whether a linear, exponential, or power model is appropriate, we make a scatter plot, a semilog plot, and a log-log plot.

- If the **scatter plot** of the data lies approximately on a line, then a linear model is appropriate.
- If the **semilog plot** of the data lies approximately on a line, then an exponential model is appropriate.
- If the **log-log plot** of the data lies approximately on a line, then a power model is appropriate.

## EXERCISES 1.5

**1.** (a) What is a one-to-one function?
(b) How can you tell from the graph of a function whether it is one-to-one?

**2.** (a) Suppose $f$ is a one-to-one function with domain $A$ and range $B$. How is the inverse function $f^{-1}$ defined? What is the domain of $f^{-1}$? What is the range of $f^{-1}$?
(b) If you are given a formula for $f$, how do you find a formula for $f^{-1}$?
(c) If you are given the graph of $f$, how do you find the graph of $f^{-1}$?

**3–14** A function is given by a table of values, a graph, a formula, or a verbal description. Determine whether it is one-to-one.

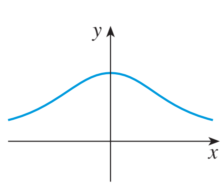
**3.**

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $f(x)$ | 1.5 | 2.0 | 3.6 | 5.3 | 2.8 | 2.0 |

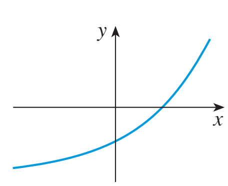
**4.**

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $f(x)$ | 1.0 | 1.9 | 2.8 | 3.5 | 3.1 | 2.9 |

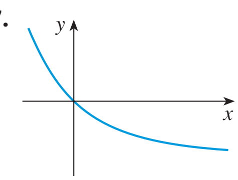
**5.**



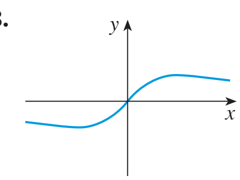

**6.**




**7.**




**8.**




**9.** $f(x) = x^2 - 2x$

**10.** $f(x) = 10 - 3x$

**11.** $g(x) = 1/x$

**12.** $g(x) = \cos x$

**13.** $f(t)$ is the height of a football $t$ seconds after kickoff.

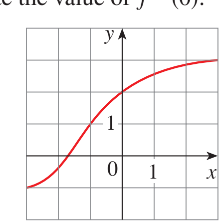
**14.** $f(t)$ is your height at age $t$.

**15.** Assume that $f$ is a one-to-one function.
(a) If $f(6) = 17$, what is $f^{-1}(17)$?
(b) If $f^{-1}(3) = 2$, what is $f(2)$?

**16.** If $f(x) = x^5 + x^3 + x$, find $f^{-1}(3)$ and $f(f^{-1}(2))$.

**17.** If $g(x) = 3 + x + e^x$, find $g^{-1}(4)$.

**18.** The graph of $f$ is given.
(a) Why is $f$ one-to-one?
(b) What are the domain and range of $f^{-1}$?
(c) What is the value of $f^{-1}(2)$?
(d) Estimate the value of $f^{-1}(0)$.



**19.** The formula $C = \frac{5}{9}(F - 32)$, where $F \geqslant -459.67$, expresses the Celsius temperature $C$ as a function of the Fahrenheit temperature $F$. Find a formula for the inverse function and interpret it. What is the domain of the inverse function?

**20.** In the theory of relativity, the mass of a particle with speed $v$ is

$$m = f(v) = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$

where $m_0$ is the rest mass of the particle and $c$ is the speed of light in a vacuum. Find the inverse function of $f$ and explain its meaning.

**21–26** Find a formula for the inverse of the function.

**21.** $f(x) = 1 + \sqrt{2 + 3x}$ **22.** $f(x) = \dfrac{4x - 1}{2x + 3}$

**23.** $f(x) = e^{2x-1}$ **24.** $y = x^2 - x, \quad x \geqslant \frac{1}{2}$

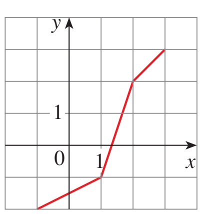
**25.** $y = \ln(x + 3)$ **26.** $y = \dfrac{e^x}{1 + 2e^x}$

**27–28** Find an explicit formula for $f^{-1}$ and use it to graph $f^{-1}$, $f$, and the line $y = x$ on the same screen. To check your work, see whether the graphs of $f$ and $f^{-1}$ are reflections about the line.

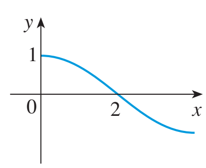
**27.** $f(x) = x^4 + 1, \quad x \geqslant 0$ **28.** $f(x) = 2 - e^x$

**29–30** Use the given graph of $f$ to sketch the graph of $f^{-1}$.

**29.**




**30.**




**31.** Let $f(x) = \sqrt{1 - x^2}, \ 0 \leqslant x \leqslant 1$.
(a) Find $f^{-1}$. How is it related to $f$?
(b) Identify the graph of $f$ and explain your answer to part (a).

**32. Vaccination coverage** Suppose we modify the function in Example 5 by introducing vaccination to control the probability of an outbreak of the disease. We want to know the fraction of the population that we have to vaccinate to achieve a target outbreak probability. If $v$ is the vaccination fraction, then the outbreak probability as a function of $v$ is

$$P = 1 - \frac{1}{R_0(1 - v)}$$

Find the inverse of this function to obtain the vaccination coverage needed for any given target outbreak probability. What do you notice about the inverse function in relation to the original function?

**33.** (a) How is the logarithmic function $y = \log_b x$ defined?
(b) What is the domain of this function?
(c) What is the range of this function?
(d) Sketch the general shape of the graph of the function $y = \log_b x$ if $b > 1$.

**34.** (a) What is the natural logarithm?
(b) What is the common logarithm?
(c) Sketch the graphs of the natural logarithm function and the natural exponential function with a common set of axes.

**35–38** Find the exact value of each expression.

**35.** (a) $\log_5 125$ (b) $\log_3\left(\frac{1}{27}\right)$

**36.** (a) $\ln(1/e)$ (b) $\log_{10}\sqrt{10}$

**37.** (a) $\log_2 6 - \log_2 15 + \log_2 20$
(b) $\log_3 100 - \log_3 18 - \log_3 50$

**38.** (a) $e^{-2\ln 5}$ (b) $\ln\left(\ln e^{e^{10}}\right)$

**39–41** Express the given quantity as a single logarithm.

**39.** $\ln 5 + 5\ln 3$

**40.** $\ln(a + b) + \ln(a - b) - 2\ln c$

**41.** $\frac{1}{3}\ln(x + 2)^3 + \frac{1}{2}[\ln x - \ln(x^2 + 3x + 2)^2]$

**42.** Use Formula 10 to evaluate each logarithm correct to six decimal places.
(a) $\log_{12} 10$ (b) $\log_2 8.4$

**43.** Suppose that the graph of $y = \log_2 x$ is drawn on a coordinate grid where the unit of measurement is an inch. How many miles to the right of the origin do we have to move before the height of the curve reaches 3 ft?

**44.** Compare the functions $f(x) = x^{0.1}$ and $g(x) = \ln x$ by graphing both $f$ and $g$ in several viewing rectangles. When does the graph of $f$ finally surpass the graph of $g$?

**45–46** Make a rough sketch of the graph of each function. Do not use a calculator. Just use the graphs given in Figures 12 and 13 and, if necessary, the transformations of Section 1.3.

**45.** (a) $y = \log_{10}(x + 5)$ (b) $y = -\ln x$

**46.** (a) $y = \ln(-x)$ (b) $y = \ln|x|$

**47–50** Solve each equation for $x$.

**47.** (a) $e^{7-4x} = 6$ (b) $\ln(3x - 10) = 2$

**48.** (a) $\ln(x^2 - 1) = 3$ (b) $e^{2x} - 3e^x + 2 = 0$

**49.** (a) $2^{x-5} = 3$ (b) $\ln x + \ln(x - 1) = 1$

**50.** (a) $\ln(\ln x) = 1$ (b) $e^{ax} = Ce^{bx}$, where $a \neq b$

**51–52** Solve each inequality for $x$.

**51.** (a) $\ln x < 0$ (b) $e^x > 5$

**52.** (a) $1 < e^{3x-1} < 2$ (b) $1 - 2\ln x < 3$

**53. Dialysis time** Hemodialysis is a process by which a machine is used to filter urea and other waste products from an individual's blood when the kidneys fail. The concentration of urea in the blood is often modeled as exponential decay. If $K$ is the mass transfer coefficient (in mL/min), $c(t)$ is the urea concentration in the blood at time $t$ (in mg/mL) and $V$ is the blood volume, then $c(t) = c_0 e^{-Kt/V}$ where $c_0$ is the initial concentration at time $t = 0$.

(a) How long should a patient be put on dialysis to reduce the blood urea concentration from an initial value of 1.65 mg/mL to 0.60 mg/mL, given that $K = 340$ mL/min and $V = 32{,}941$ mL?

(b) Derive a general formula for the dialysis time $T$ in terms of the initial urea concentration $c_0$ and the target urea concentration $c(T)$.

**54. Dialysis treatment adequacy**

(a) The quantity $Kt/V$ in Exercise 53 is sometimes used as a measure of dialysis treatment adequacy. What does this represent and what are its units?

(b) Another quantity sometimes used to measure dialysis treatment adequacy is the fractional reduction in urea during a dialysis session, denoted by $U$ (that is, the ratio of the amount of urea removed during dialysis to its initial amount). What is the relationship between $U$ and $Kt/V$?

**55.** (a) Find the domain of $f(x) = \ln(e^x - 3)$.
(b) Find $f^{-1}$ and its domain.

**56.** (a) What are the values of $e^{\ln 300}$ and $\ln(e^{300})$?
(b) Use your calculator to evaluate $e^{\ln 300}$ and $\ln(e^{300})$. What do you notice? Can you explain why the calculator has trouble?

**57. Bacteria population** If a bacteria population starts with 500 bacteria and doubles in size every half hour, then the number of bacteria after $t$ hours is $n = f(t) = 500 \cdot 4^t$. (See Exercise 1.4.30).

(a) Find the inverse of this function and explain its meaning.

(b) When will the population reach 10,000?

**58.** When a camera flash goes off, the batteries immediately begin to recharge the flash's capacitor, which stores electric charge given by

$$Q(t) = Q_0(1 - e^{-t/a})$$

(The maximum charge capacity is $Q_0$ and $t$ is measured in seconds.)

(a) Find the inverse of this function and explain its meaning.

(b) How long does it take to recharge the capacitor to 90% of capacity if $a = 2$?

**59–64** Data points $(x, y)$ are given.

(a) Draw a scatter plot of the data points.

(b) Make semilog and log-log plots of the data.

(c) Is a linear, power, or exponential function appropriate for modeling these data?

(d) Find an appropriate model for the data and then graph the model together with a scatter plot of the data.

**59.**

| | | | | | | |
|---|---|---|---|---|---|---|
| $x$ | 2 | 4 | 6 | 8 | 10 | 12 |
| $y$ | 0.08 | 0.12 | 0.18 | 0.26 | 0.35 | 0.53 |

**60.**

| | | | | | | |
|---|---|---|---|---|---|---|
| $x$ | 1.0 | 2.4 | 3.1 | 3.6 | 4.3 | 4.9 |
| $y$ | 3.2 | 4.8 | 5.8 | 6.2 | 7.2 | 7.9 |

**61.**

| | | | | | | |
|---|---|---|---|---|---|---|
| $x$ | 0.5 | 1.0 | 1.5 | 2.0 | 2.5 | 3.0 |
| $y$ | 4.10 | 3.71 | 3.39 | 3.2 | 2.78 | 2.53 |

**62.**

| | | | | | | |
|---|---|---|---|---|---|---|
| $x$ | 10 | 20 | 30 | 40 | 50 | 60 |
| $y$ | 29 | 82 | 150 | 236 | 330 | 430 |

**63.**

| | | | | | | |
|---|---|---|---|---|---|---|
| $x$ | 3 | 4 | 5 | 6 | 7 | 8 |
| $y$ | 11.3 | 20.2 | 32.2 | 45.7 | 62.1 | 80.4 |

**64.**

| | | | | | | |
|---|---|---|---|---|---|---|
| $x$ | 5 | 10 | 15 | 20 | 25 | 30 |
| $y$ | 0.013 | 0.046 | 0.208 | 0.930 | 4.131 | 18.002 |

**65. Indian population** The table gives the midyear population of India (in millions) for the last half of the 20th century.

| Year | Population | Year | Population |
|---|---|---|---|
| 1950 | 370 | 1980 | 685 |
| 1960 | 445 | 1990 | 838 |
| 1970 | 554 | 2000 | 1006 |

(a) Make a scatter plot, semilog plot, and log-log plot for these data and comment on which type of model would be most appropriate.

(b) Obtain an exponential model for the population.
(c) Use your model to estimate the population in 2010 and compare with the actual population of 1173 million. What conclusion can you make?

**66. Why is the dodo extinct?** BB Ornithologists measured and cataloged the wingspans and weights of many different species of birds that can fly. The table shows the wingspan $L$ for a bird of weight $W$.
(a) Make a scatter plot, semilog plot, and log-log plot for these data. Which type of model do you think would be best?
(b) Find an exponential model and power model for the data.
(c) Graph the models from part (b). Which is better?
(d) The dodo is a bird that has been extinct since the late 17th century. It weighed about 45 pounds and had a wingspan of about 20 inches.Use the model chosen in part (c) to estimate the wingspan of a 45-lb bird. Why couldn't a dodo fly?

| Bird | $W$ (lb) | $L$ (in) |
|---|---|---|
| Turkey vulture | 4.40 | 69 |
| Bald eagle | 6.82 | 84 |
| Great horned owl | 3.08 | 44 |
| Cooper's hawk | 1.03 | 28 |
| Sandhill crane | 9.02 | 79 |
| Atlantic puffin | 0.95 | 24 |
| California condor | 17.82 | 109 |
| Common loon | 7.04 | 48 |
| Yellow warbler | 0.022 | 8 |
| Common grackle | 0.20 | 16 |
| Wood stork | 5.06 | 63 |
| Mallard | 2.42 | 35 |


Study of a Dodo (oil on canvas), Hart, F. (19th Century)/Royal Albert Memorial Museum, Exeter, Devon, UK/The Bridgeman Art Lbrary.

The dodo (now extinct)

**67. Biodiversity in a rain forest** To quantify the biodiversity of trees in a tropical rain forest, biologists collected data in the Pasoh Forest Reserve of Malaysia. The table shows the number of tree species $S$ found for a given area $A$ in the rain forest.
(a) Make a scatter plot of the data and either a semilog plot or a log-log plot, whichever you think is more appropriate.
(b) Use your preferred plot from part (a) to find a model. Graph your model with the scatter plot.

| $A$ (m$^2$) | $S$ | $A$ (m$^2$) | $S$ |
|---|---|---|---|
| 3.81 | 3 | 122.07 | 70 |
| 7.63 | 3 | 244.14 | 112 |
| 15.26 | 12 | 488.28 | 134 |
| 30.52 | 13 | 976.56 | 236 |
| 61.04 | 31 | | |

*Source:* Adapted from K. Kochummen et al., "Floristic Composition of Pasoh Forest Reserve, a Lowland Rain Forest in Peninsular Malaysia," *Journal of Tropical Forest Science* 3 (1991): 1–13.

**68. Malarial parasites** The table, supplied by Andrew Read, shows the results of an experiment involving malarial parasites. The time $t$ is measured in days and $N$ is the number of parasites per microliter of blood.
(a) Make a scatter plot and a semilog plot of the data.
(b) Find an exponential model and graph your model with the scatter plot. Is it a good fit?

| $t$ (days) | $N$ | $t$ (days) | $N$ |
|---|---|---|---|
| 1 | 228 | 5 | 372,331 |
| 2 | 2,357 | 6 | 2,217,441 |
| 3 | 12,750 | 7 | 6,748,400 |
| 4 | 26,661 | | |

**69. Drinking and driving** In a medical study, researchers measured the mean blood alcohol concentration (BAC) of eight fasting adult male subjects (in mg/mL) after rapid consumption of 30 mL of ethanol (corresponding to two standard alcoholic drinks). The BAC peaked after half an hour and the table shows measurements starting after an hour.

| $t$ (hours) | 1.0 | 1.25 | 1.5 | 1.75 | 2.0 |
|---|---|---|---|---|---|
| BAC | 0.33 | 0.29 | 0.24 | 0.22 | 0.18 |

| $t$ (hours) | 2.25 | 2.5 | 3.0 | 3.5 | 4.0 |
|---|---|---|---|---|---|
| BAC | 0.15 | 0.12 | 0.069 | 0.034 | 0.010 |

(a) Make a scatter plot and a semilog plot of the data.
(b) Find an exponential model and graph your model with the scatter plot. Is it a good fit?

(c) Use your model and logarithms to determine when the BAC will be less than 0.08 mg/mL, the legal limit for driving.

*Source:* Adapted from P. Wilkinson et al., "Pharmacokinetics of Ethanol after Oral Administration in the Fasting State," *Journal of Pharmacokinetics and Biopharmaceutics* 5 (1977): 207–24.

**70. Amplifying DNA** Polymerase Chain Reaction (PCR) is a biochemical technique that allows scientists to take tiny samples of DNA and amplify them into large samples that can then be examined to determine the DNA sequence. (This is useful, for example, in forensic science.) The process works by mixing the sample with appropriate enzymes and then heating it until the DNA double helix separates into two individual strands. The enzymes then copy each strand, and once the sample is cooled the number of DNA molecules will have doubled. By repeatedly performing this heating and cooling process, the number of DNA molecules continues to double every temperature cycle (referred to as a PCR cycle).

(a) Suppose a sample containg $x$ molecules is collected from a crime scene and is amplified by PCR. Express the number of DNA molecules as a function of the number $n$ of PCR cycles.

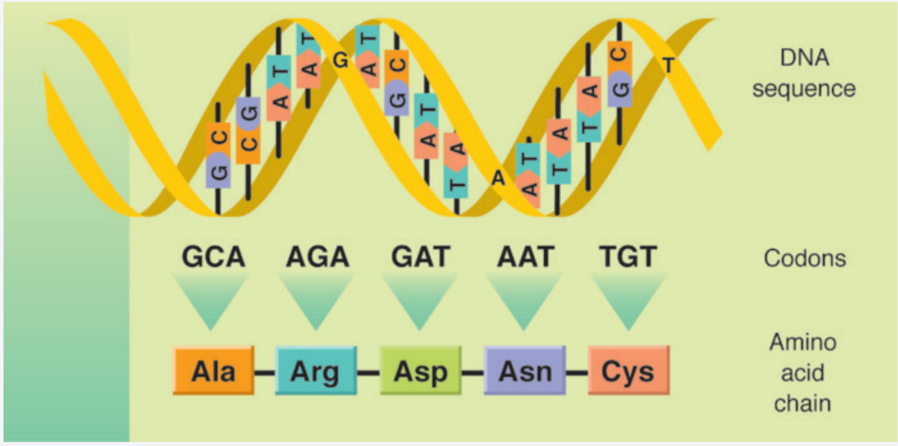
(b) There is a detection threshold of $T$ molecules below which no DNA can be seen. Derive an equation for the number of PCR cycles it will take for the DNA sample to reach the detection threshold.

(c) One way scientists determine the abundance of different DNA molecules in a sample is by measuring the difference in time it takes to reach the detection threshold for each. Sketch a graph of the number of cycles needed to reach the detection threshold as a function of the initial number of molecules. Comment on the relationship between differences in initial number of molecules and differences in the time to reach the detection threshold.

## PROJECT The Coding Function of DNA

BB

Proteins are made up of long chains of molecules called amino acids. Twenty different amino acids are coded by the DNA of living organisms. The "alphabet" of DNA consists of four letters A, T, C, and G, called *bases*. These bases are grouped together along the DNA sequence into "words," called *codons*. All codons contain the same number of bases (that is, the words are always the same length) and any given codon specifies exactly one amino acid (see Figure 1). The coding of amino acids by DNA described here can be viewed as a *function* that takes an input codon and produces an output amino acid.




1. Suppose that all codons contained only one base. What would the domain of the coding function be? What would its biggest possible range be?

2. Suppose that all codons contained two bases. What would the domain of the coding function be? What would its biggest possible range be?

3. In fact, all codons actually contain three bases. What is the domain of the coding function in this case? What is its biggest possible range?

4. Given your answers to Problems 1–3, speculate on why codons contain three bases rather than fewer or more.

5. Explain why the coding function in which codons have three bases is not a one-to-one function.

Researchers use the fact that the coding function is not one-to-one to infer which DNA sequence variants are advantageous. In particular, because the coding function is not one-to-one, two kinds of DNA mutations can occur: those that do not alter the amino acid (called synonymous mutations) and those that do (called nonsynonymous mutations). Synonymous mutations are not expected to affect organismal functioning because they don't affect protein structure. As a result, such synonymous mutations are expected to accumulate over time, by chance, in a clocklike fashion. Nonsynonymous mutations do change the amino acid and therefore alter protein structure. If such alterations are advantageous, then we would expect these mutational changes to occur at a rate that is higher than those of the neutral, synonymous mutations. This kind of comparison is possible only because the genetic coding function isn't one-to-one; it forms the basis of nonsynonymous-to-synonymous ratio tests used in biology.

## 1.6 Sequences and Difference Equations

A **sequence** can be thought of as a list of numbers written in a definite order:

$$a_1,\ a_2,\ a_3,\ a_4,\ \ldots,\ a_n,\ \ldots$$

The number $a_1$ is called the *first term*, $a_2$ is the *second term*, and in general $a_n$ is the *nth term*. Sometimes we might want to start the sequence with $n = 0$. Then $a_0$ is the zeroth term and we list the terms of the sequence as

$$a_0,\ a_1,\ a_2,\ a_3\ ,\ldots,\ a_n,\ \ldots$$

Notice that for every positive integer $n$ there is a corresponding number $a_n$ and so a sequence can be defined as a function whose domain is the set of positive integers. But we usually write $a_n$ instead of the function notation $f(n)$ for the value of the function at the number $n$.

Some sequences are defined by giving a formula for the $n$th term $a_n$ in terms of $n$, as the following example illustrates.

**EXAMPLE 1** | Find the first five terms of the sequence.

(a) $a_n = \dfrac{n}{n+1}, \quad n \geqslant 1$ (b) $a_n = (-1)^{n-1}, \quad n \geqslant 1$ (c) $a_n = 8^n, \quad n \geqslant 0$

**SOLUTION**

(a) Putting $n = 1, 2, 3, 4, 5$, successively, in the formula for $a_n$ we get the initial terms of the sequence:

$$\frac{1}{2},\ \frac{2}{3},\ \frac{3}{4},\ \frac{4}{5},\ \frac{5}{6},\ \ldots,\ \frac{n}{n+1},\ \ldots$$

(b) Again we start with $n = 1$, noting that $(-1)^{1-1} = (-1)^0 = 1$:

$$1, \ -1, \ 1, \ -1, \ 1, \ \ldots$$

(c) Here we start with $n = 0$. The terms are $8^0, 8^1, 8^2, 8^3, 8^4$, or

$$1, \ 8, \ 64, \ 512, \ 4096, \ \ldots$$

We've already met this sequence in describing the growth of malarial parasites in Section 1.4. ■

A sequence can be pictured by plotting its **graph**. Because a sequence is a function whose domain is the set of positive integers, its graph consists of isolated points with coordinates

$$(1, a_1) \qquad (2, a_2) \qquad (3, a_3) \qquad \ldots \qquad (n, a_n) \qquad \ldots$$

Parts (a) and (b) of Figure 1 show the graphs of the sequences in parts (a) and (b) of Example 1.

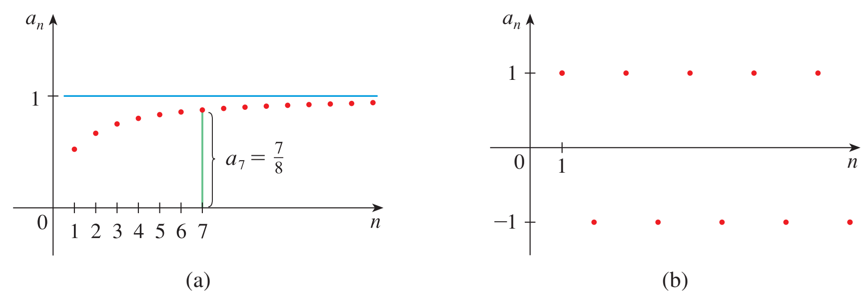



**FIGURE 1**

The heights of the points on the graph in Figure 1(a) appear to be approaching the number 1, whereas those in Figure 1(b) are oscillating (forever) between $-1$ and 1. The behavior of sequences in the long run (as $n$ becomes large) will be discussed in Section 2.1.

If we know the first few terms of a sequence but don't have a general formula for $a_n$, we might be able to detect a pattern in the numbers and write a formula for the $n$th term. Such a formula might not be unique; we look for the simplest formula, as in the next example.

**EXAMPLE 2** | Find a formula for the general term $a_n$ of the sequence

$$\frac{3}{5}, \ -\frac{4}{25}, \ \frac{5}{125}, \ -\frac{6}{625}, \ \frac{7}{3125}, \ \ldots$$

assuming that the pattern of the first few terms continues.

**SOLUTION** We are given that

$$a_1 = \frac{3}{5} \qquad a_2 = -\frac{4}{25} \qquad a_3 = \frac{5}{125} \qquad a_4 = -\frac{6}{625} \qquad a_5 = \frac{7}{3125}$$

Notice that the numerators of these fractions start with 3 and increase by 1 whenever we go to next term. The second term has numerator 4, the third term has numerator 5; in general, the $n$th term will have numerator $n + 2$. The denominators are the powers of 5, so $a_n$ has denominator $5^n$. The signs of the terms are alternately positive and negative, so we multiply by $(-1)^{n-1}$ as in Example 1(b). Therefore

$$a_n = (-1)^{n-1}\frac{n+2}{5^n}$$

■

## ■ Recursive Sequences: Difference Equations

Some sequences don't have simple defining formulas like the ones in Examples 1 and 2. The *n*th term of a sequence may depend on some of the terms that precede it. A sequence defined in this way is called **recursive**.

**EXAMPLE 3** | Find the first five terms of the sequence defined recursively by the equations

$$a_1 = 2 \qquad a_{n+1} = \tfrac{1}{2}(a_n + 6) \qquad \text{for } n \geqslant 1$$

**SOLUTION** The defining formula allows us to calculate a term if we know the preceding one. We are given the first term, so we can use it to find the second term. Then we find the third term from the second one, and so on:

$$a_2 = \tfrac{1}{2}(a_1 + 6) = \tfrac{1}{2}(2 + 6) = 4$$

$$a_3 = \tfrac{1}{2}(a_2 + 6) = \tfrac{1}{2}(4 + 6) = 5$$

$$a_4 = \tfrac{1}{2}(a_3 + 6) = \tfrac{1}{2}(5 + 6) = 5.5$$

$$a_5 = \tfrac{1}{2}(a_4 + 6) = \tfrac{1}{2}(5.5 + 6) = 5.75$$

So the sequence starts like this:

$$2, \ 4, \ 5, \ 5.5, \ 5.75, \ \ldots$$

■

**EXAMPLE 4** | Fibonacci sequence Find the first ten terms of the recursive sequence given by

$$F_1 = 1, \qquad F_2 = 1, \qquad F_n = F_{n-1} + F_{n-2} \qquad \text{for } n \geqslant 3$$

**SOLUTION** We are given $F_1$ and $F_2$, so we proceed as follows:

$$F_3 = F_2 + F_1 = 1 + 1 = 2$$

$$F_4 = F_3 + F_2 = 2 + 1 = 3$$

$$F_5 = F_4 + F_3 = 3 + 2 = 5$$

Each term is the sum of the two terms that precede it, so we can easily write as many terms as we please. Here are the first ten terms:

$$1, \ 1, \ 2, \ 3, \ 5, \ 8, \ 13, \ 21, \ 34, \ 55$$

■

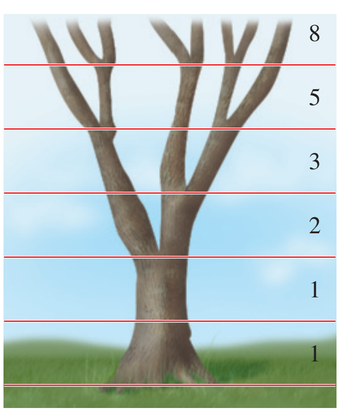



**FIGURE 2**
The Fibonacci sequence in the branching of a tree

The sequence in Example 4 is called the **Fibonacci sequence**, named after the 13th-century Italian mathematician known as Fibonacci, who used it to solve a problem concerning the breeding of rabbits (see Exercise 23). This sequence also occurs in numerous applications in plant biology. (See Figure 2 for one such occurrence.)

A recursive sequence is also called a **difference equation**. The recursive sequence in Example 3 is called a **first-order difference equation** because $a_{n+1}$ depends on just the preceding term $a_n$, whereas the Fibonacci sequence is a **second-order difference equation** because $F_n$ depends on the two preceding terms $F_{n-1}$ and $F_{n-2}$.

The general first-order difference equation is of the form

$$a_{n+1} = f(a_n)$$

where $f$ is some function. Why is it called a *difference equation*? The word *difference* comes from the fact that such equations are often formulated in terms of the difference between one term and the next:

$$\Delta a_n = a_{n+1} - a_n$$

The equation $\Delta a_n = g(a_n)$ can be written as follows:

$$a_{n+1} - a_n = g(a_n)$$
$$a_{n+1} = a_n + g(a_n) = f(a_n)$$

where $f(x) = x + g(x)$.

BB

## ■ Discrete-Time Models in the Life Sciences

Difference equations are often used in biology to model cell division and insect populations, for example. In these contexts we usually replace $n$ by $t$, to denote time. If we think of $t$ as the current time, then $t + 1$ is one unit of time into the future. (For cell division, $t$ might represent hours or days; for insect populations, it could represent days, months, or years.) We will use $N_t$ to denote the population size, so a difference equation modeling population size has the form

$$N_{t+1} = f(N_t) \qquad t = 0, 1, 2, 3, \ldots$$

In this context we call $f$ an **updating function** because $f$ "updates" the population from $N_t$ to $N_{t+1}$.

We have already seen an example of this in Section 1.4, where a malarial parasite produces 8 new parasites in a period of 24 hours. So

$$N_{t+1} = 8N_t \qquad N_0 = 1$$

where $t$ is measured in days, and we saw that the solution of this difference equation is

$$N_t = 8^t$$

If an *E. coli* population starts with $N_0$ bacteria and its size doubles every 20 minutes, then we measure $t$ in units of 20 minutes and write $N_{t+1} = 2N_t$. As before, we find that

$$N_t = N_0 \cdot 2^t$$

More generally, if a population of cells divides synchronously, with each cell producing $R$ daughter cells, then the difference equation

We have discussed sequences defined by a formula and also recursive sequences. Equation 1 defines a recursive sequence and the solution given by Equation 2 is the corresponding formula.

**(1)** $$\boxed{N_{t+1} = RN_t}$$

relates successive generations and the solution is

**(2)** $$N_t = N_0 \cdot R^t$$

The number $R$ is the number of offspring per individual and is called the **per capita growth factor**.

Similar equations arise when we consider insect populations that breed seasonally. We will take the unit of time to be the time span from one generation to the next. Then in general terms we can formulate the model

$$N_{t+1} = N_t + \text{inflow} - \text{outflow} \tag{3}$$

If the population has a constant per capita birth rate $\beta$ and constant per capita death rate $\mu$, then the difference equation in (3) becomes

$$N_{t+1} = N_t + \beta N_t - \mu N_t \tag{4}$$

Notice that in the case where insects die immediately after producing the next generation, we have $\mu = 1$ and Equation 4 becomes Equation 1 with $R = \beta$.

So far we have considered populations that grow under ideal conditions without limitations to growth. Let's now consider a more realistic model called the **logistic difference equation**. Let $K$ represent the **carrying capacity**, which is the population size at which the per capita growth factor is 1. We replace the difference equation in Equation 1, $N_{t+1} = RN_t$, by the model

$$\boxed{N_{t+1} = \left[1 + r\left(1 - \frac{N_t}{K}\right)\right]N_t} \tag{5}$$

where $r$ is a positive constant. Here the per capita growth factor is

$$\boxed{R = 1 + r\left(1 - \frac{N_t}{K}\right)} \tag{6}$$

whereas in (1) $R$ is simply a constant. Notice that $N_t/K$ is the fraction of the carrying capacity at time $t$ and so $r(1 - N_t/K)$ is small when $N_t$ is close to $K$ and $r(1 - N_t/K)$ is close to its largest value $r$ when $N_t$ is small. Observe that $R$ decreases linearly from $1 + r$ to 1 as $N_t$ increases from 0 to $K$. This means that the logistic difference equation has a variable per capita growth factor $R$.

We can simplify Equation 5 by defining

$$x_t = \frac{r}{(1 + r)K}N_t$$

Then

$$x_{t+1} = \frac{r}{(1 + r)K}N_{t+1} = \frac{r}{(1 + r)K}\left[1 + r\left(1 - \frac{N_t}{K}\right)\right]N_t$$

$$= \frac{r}{(1 + r)K}\left[(1 + r)N_t - \frac{rN_t^2}{K}\right] = \frac{r}{K}N_t - \frac{r^2N_t^2}{(1 + r)K^2}$$

On the right side of this equation we use the fact that

$$N_t = \frac{(1 + r)K}{r}x_t$$

to obtain $\qquad x_{t+1} = (1 + r)x_t - (1 + r)x_t^2 = (1 + r)x_t(1 - x_t)$

If we now write $R_{\max} = 1 + r$, which is the largest value of $R$ as a function of $N_t$ in Equation 6, we obtain a simpler-looking difference equation:

**(7)** $$\boxed{x_{t+1} = R_{\max} x_t(1 - x_t)}$$

Equation 7 is also called the **logistic difference equation**.

**EXAMPLE 5** | **BB** If $x_0 = \frac{7}{8}$, graph the first ten terms of the logistic difference equation (7) for (a) $R_{\max} = 1.5$ and (b) $R_{\max} = 3.2$.

**SOLUTION**

(a) With $x_0 = \frac{7}{8}$ and $x_{t+1} = 1.5x_t(1 - x_t)$, we use a graphing calculator or computer to calculate the first ten terms approximately and then we plot them in Figure 3.

| $t$ | $x_t$ | $t$ | $x_t$ |
|---|---|---|---|
| 0 | 0.8750 | 6 | 0.3155 |
| 1 | 0.1641 | 7 | 0.3239 |
| 2 | 0.2057 | 8 | 0.3285 |
| 3 | 0.2451 | 9 | 0.3309 |
| 4 | 0.2775 | 10 | 0.3321 |
| 5 | 0.3008 | | |

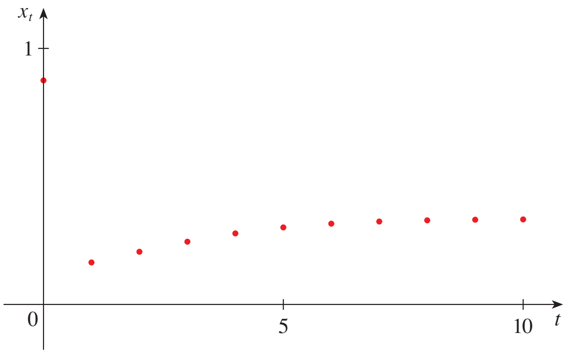



**FIGURE 3**

(b) With $R_{\max} = 3.2$ we get the following values and plot them in Figure 4.

| $t$ | $x_t$ | $t$ | $x_t$ |
|---|---|---|---|
| 0 | 0.8750 | 6 | 0.7604 |
| 1 | 0.3500 | 7 | 0.5831 |
| 2 | 0.7280 | 8 | 0.7779 |
| 3 | 0.6337 | 9 | 0.5528 |
| 4 | 0.7428 | 10 | 0.7911 |
| 5 | 0.6113 | | |

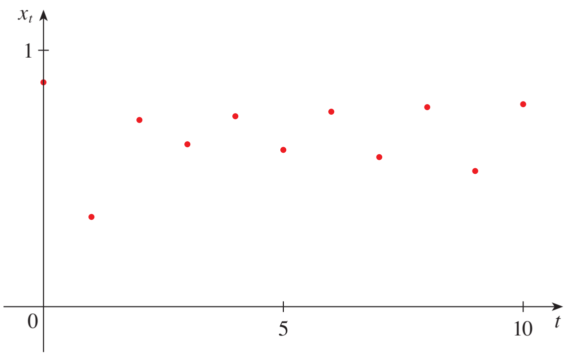



**FIGURE 4** ■

Notice from Figures 3 and 4 that when we change the value of $r$ in the logistic difference equation, and thereby change $R_{\max}$, the sequence looks quite different. We will return to this equation in Sections 2.1 and 4.5 to explore the limiting behavior of the logistic difference equation for different values of $R_{\max}$.

**EXAMPLE 6** | **BB** **Maintaining cerebrospinal pressure**[1] Cerebrospinal fluid (CSF) is a clear liquid that occupies the compartment of the body containing the

1. Adapted from S. Cruickshank, *Mathematics and Statistics in Anaesthesia* (New York: Oxford University Press, USA, 1998).

brain and spinal cord (the cerebrospinal chamber). It is important to maintain an appropriate cerebrospinal pressure during medical procedures and the pressure is a function of the CSF volume. Suppose that we measure the CSF volume every five minutes during a medical procedure and that $A$ mL of CSF is secreted into the cerebrospinal chamber every five minutes. Also, suppose that the amount of CSF reabsorbed every five minutes is proportional to its current volume.

(a) Derive a discrete-time difference equation for how the volume $V$ of CSF changes from one measurement to the next.

(b) Ultimately we are interested in how the cerebrospinal pressure $P$ changes from one measurement to the next. Suppose that pressure is related to volume according to the equation $P = V^2$. (This would be appropriate if pressure is nearly zero for small volumes but increases at an accelerating rate as volume increases.) Derive a discrete-time recursion for the pressure.

**SOLUTION**

(a) The volume at measurement $m + 1$ is the volume at measurement $m$ plus the secreted CSF ($A$) minus what is reabsorbed ($kV$, where $k$ is the constant of proportionality). Therefore we have

$$V_{m+1} = V_m + A - kV_m = A + (1 - k)V_m$$

(b) For any given measurement, the pressure is given by $P_m = V_m^2$. So

$$P_{m+1} = V_{m+1}^2 = [A + (1 - k)V_m]^2$$

This recursion is not yet complete because it tells us the pressure at measurement $m + 1$ as a function of the *volume* at measurement $m$. To use it recursively we need the recursion to give us the pressure at measurement $m + 1$ as a function of the *pressure* at measurement $m$. Therefore we need to write $V_m$ in terms of $P_m$. Solving $P_m = V_m^2$ for $V_m$ and keeping only the positive solution, we get $V_m = \sqrt{P_m}$. Substituting, we get

$$\begin{aligned} P_{m+1} &= \left[A + (1 - k)\sqrt{P_m}\right]^2 \\ &= A^2 + 2A(1 - k)\sqrt{P_m} + (1 - k)^2 P_m \end{aligned}$$ ■

## EXERCISES 1.6

**1–4** List the first five terms of the sequence.

**1.** $a_n = \dfrac{2n}{n^2 + 1}$

**2.** $a_n = \dfrac{3^n}{1 + 2^n}$

**3.** $a_n = \dfrac{(-1)^{n-1}}{5^n}$

**4.** $a_n = \cos \dfrac{n\pi}{2}$

**5–8** Calculate, to four decimal places, the first ten terms of the sequence and use them to plot the graph of the sequence by hand.

**5.** $a_n = \dfrac{3n}{1 + 6n}$

**6.** $a_n = 2 + \dfrac{(-1)^n}{n}$

**7.** $a_n = 1 + \left(-\frac{1}{2}\right)^n$

**8.** $a_n = 1 + \dfrac{10^n}{9^n}$

**9–14** Find a formula for the general term $a_n$ of the sequence, assuming that the pattern of the first few terms continues.

**9.** $1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \frac{1}{9}, \ldots$

**10.** $1, -\frac{1}{3}, \frac{1}{9}, -\frac{1}{27}, \frac{1}{81}, \ldots$

**11.** $-3, 2, -\frac{4}{3}, \frac{8}{9}, -\frac{16}{27}, \ldots$

**12.** $5, 8, 11, 14, 17, \ldots$

**13.** $\frac{1}{2}, -\frac{4}{3}, \frac{9}{4}, -\frac{16}{5}, \frac{25}{6}, \ldots$

**14.** $1, 0, -1, 0, 1, 0, -1, 0, \ldots$

**15–22** Find the first six terms of the recursive sequence.

**15.** $a_1 = 1, a_{n+1} = 5a_n - 3$

**16.** $a_1 = 6, a_{n+1} = \dfrac{a_n}{n}$

**17.** $a_1 = 2, a_{n+1} = \dfrac{a_n}{1 + a_n}$

**18.** $a_1 = 1, a_{n+1} = 4 - a_n$

**19.** $a_1 = 1, a_{n+1} = \sqrt{3a_n}$

**20.** $a_1 = 3, a_{n+1} = \sqrt{3a_n}$

**21.** $a_1 = 2, a_2 = 1, a_{n+1} = a_n - a_{n-1}$

**22.** $a_1 = 1, a_2 = 2, a_{n+2} = a_{n+1} + 2a_n$

**23. Breeding rabbits** Fibonacci posed the following problem: Suppose that rabbits live forever and that every month each pair produces a new pair, which becomes reproductive at age 2 months. If we start with one newborn pair, how many pairs of rabbits will we have in the $n$th month? Show that the answer is $F_n$, the $n$th term of the Fibonacci sequence defined in Example 4.

**24. Harvesting fish** A fish farmer has 5000 catfish in his pond. The number of catfish increases by 8% per month and the farmer harvests 300 catfish per month.

(a) Show that the catfish population $P_n$ after $n$ months is given recursively by

$$P_n = 1.08P_{n-1} - 300 \qquad P_0 = 5000$$

(b) How many catfish are in the pond after six months?

**25.** Consider the difference equation

$$N_0 = 1 \qquad N_{t+1} = RN_t$$

What can you say about the solution of this equation as $t$ becomes large in the following three cases?

(a) $R < 1$ (b) $R = 1$ (c) $R > 1$

**26.** (a) For a difference equation of the form $N_{t+1} = f(N_t)$, calculate the composition $(f \circ f)(N_t)$. What is the meaning of $f \circ f$ in this context?

(b) If $N_{t+1} = f(N_t)$, where $f$ is one-to-one, what is $f^{-1}(N_{t+1})$? What is the meaning of the inverse function $f^{-1}$ in this context?

**27–31 Logistic equation** For the logistic difference equation $x_{t+1} = cx_t(1 - x_t)$ and the given values of $x_0$ and $c$, calculate $x_t$ to four decimal places for $t = 1, 2, \ldots, 10$ and graph $x_t$. Comment on the behavior of the sequence.

**27.** $x_0 = 0.5,\ c = 1.5$

**28.** $x_0 = 0.5,\ c = 2.5$

**29.** $x_0 = \frac{7}{8},\ c = 3.42$

**30.** $x_0 = \frac{7}{8},\ c = 3.45$

**31.** $x_0 = 0.5,\ c = 3.7$

**32. Ricker equation** In the logistic difference equation the factor $(1 - x_t)$ decreases linearly from 1 to 0 as $x_t$ increases from 0 to 1. If, instead, we introduce the decreasing exponential factor $e^{-x_t}$, we get what is called the *Ricker difference equation*:

$$x_{t+1} = cx_t e^{-x_t}$$

This model has the advantage that the factor $e^{-x_t}$ never reaches 0.

(a) If $x_0 = 0.5$ and $c = 2.5$, calculate the first ten terms to four decimal places and graph them.

(b) Compare with Exercise 28.

**33.** Repeat Exercise 32(a) for $x_0 = \frac{7}{8}$ and $c = 3.42$. Compare with Exercise 29.

**34. Drug concentration** Suppose $C_t$ is the concentration of a drug in the bloodstream at time $t$, $A$ is the concentration of the drug that is administered at each time step, and $k$ is the fraction of the drug metabolized in a time step.

(a) What is the recursion that models how the drug concentration changes?

(b) If the initial concentration is $C_0 = 120\ \mu g/mL$ and $A = 80\ \mu g/mL$ and $k = \frac{1}{2}$, plot some points on the graph of $C_t$. Is the graph similar for other values of $A$ and $k$?

**35. Bacteria colonies on agar plates** Bacteria are often grown on agar plates and form circular colonies. The area of a colony is proportional to the number of bacteria it contains. The agar (a gelatinous substance obtained from red seaweed) is the resource that bacteria use to reproduce and so only those bacteria on the edge of the colony can produce new offspring. Therefore the population changes according to the equation $N_{t+1} = N_t + I$, where $I$ is the input of new individuals and is proportional to the circumference of the colony, with proportionality constant $R$.

(a) Derive the recursion for the population size.

(b) Plot some points on the graph of $N_t$, assuming specific values for the proportionality constants.

**36. Spherical colonies** Suppose the volume of a spherical colony is proportional to the number of individuals in it and growth occurs only at the surface-resource interface of the colony. Find a difference equation that models the population.

**37. Salmon and bears** Pacific salmon populations have discrete breeding cycles in which they return from the ocean to streams to reproduce and then die. This occurs every one to five years, depending on the species.

(a) Suppose that each fish must first survive predation by bears while swimming upstream, and predation occurs with probability $d$. After swimming upstream, each fish produces $b$ offspring before dying. The stream is then stocked with $m$ additional newly hatched fish before all fish then swim out to sea. What is the discrete-time recursion for the population size, assuming that there is no mortality while at sea? You should count the population immediately before the upstream journey.

(b) Suppose instead that bears prey on fish only while the fish are swimming downstream. What is the discrete-time recursion for the population dynamics? (Again assume there is no mortality while at sea.)

(c) Which of the recursions obtained in parts (a) and (b) predicts the largest increase in population size from one year to the next? Justify your answer both mathematically and in terms of the underlying biology. You can assume that $0 < d < 1$ and $b > 0$.

**38. Methyl groups in DNA** DNA sometimes has chemical groups attached, called methyl groups, that affect gene expression. Suppose that, during each hour, first a fraction $m$ of unmethylated locations on the DNA become methylated, and then a fraction $u$ of methylated locations become unmethylated. Find a recursion for the fraction $f$ of the DNA molecule that is methylated.

**39. Two bacteria strains** Suppose the population sizes of two strains of bacteria each grow as described by the recursions $a_{t+1} = R_a a_t$ and $b_{t+1} = R_b b_t$, respectively. The frequency of the first strain at time $t$ is defined as $p_t = a_t/(a_t + b_t)$. Derive a recursion for $p_t$ and show that it can be written in terms of a single constant $\alpha = R_a/R_b$.

**40.** Find the first 40 terms of the sequence defined by

$$a_{n+1} = \begin{cases} \frac{1}{2}a_n & \text{if } a_n \text{ is an even number} \\ 3a_n + 1 & \text{if } a_n \text{ is an odd number} \end{cases}$$

and $a_1 = 11$. Do the same if $a_1 = 25$. Make a conjecture about this type of sequence.

## PROJECT Drug Resistance in Malaria

BB

Drug resistance in malaria is a serious problem in many parts of the world. Let's suppose that there are two different genes in the population, one that causes resistance (labeled R) and one that is sensitive to the drug (labeled S). We will construct a recursion for the frequency of the R gene by *assuming* that the entire malaria life cycle occurs synchronously.

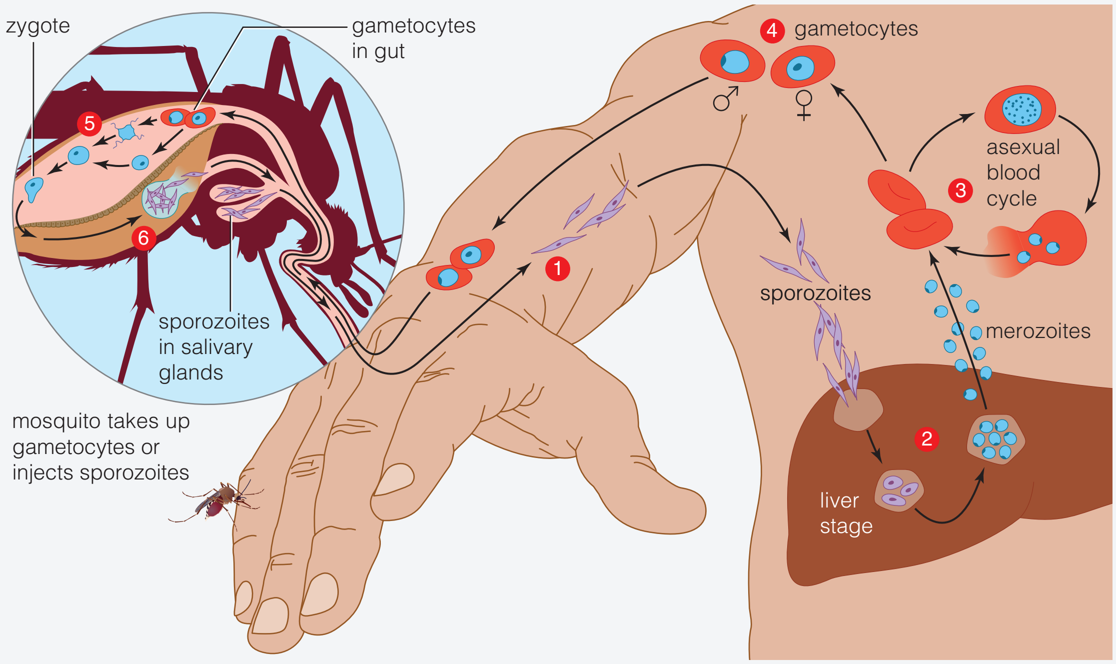
The life cycle of malaria is quite complicated. Part of it occurs in mosquitoes and part in humans (see Figure 1). While in humans, each malaria parasite carries a single copy of the gene (either R or S) and is referred to as **haploid**. But while in the mosquito, pairs of such haploid parasites combine through a process of sexual reproduction to form **diploid** parasites. Diploid individuals carry two copies of the gene and therefore can be of three different types: RR, RS, or SS (see Figure 2).

To contruct a recursion for the frequency of the R gene, we must first choose a point in the malaria life cycle at which to census the population. Since the haploid stage of the parasite in humans is the simplest, let's choose that. We use $p_t$ to denote the frequency of the R type in the haploid stage. (All R genes are colored yellow in Figure 2.) Our goal is to derive a recursion of the form $p_{t+1} = f(p_t)$ for some function $f$. We do this by dividing the entire life cycle into three steps: (1) union of pairs of haploid individuals to form diploid individuals; (2) differential survival of diploid individuals in the mosquito; and (3) the production of new haploid individuals by the surviving diploid parasites.

**1.** Suppose that when pairs of haploid individuals unite to form diploids in the mosquitoes, they do so randomly and independently of the gene they carry. If $p_t$ is the frequency of the R gene in the haploids, what are the frequencies of the RR, RS, and SS types in the mosquitoes after this step occurs? (As a check, make sure the frequencies sum to 1.)

**2.** Suppose that the probability of survival of the three different types of diploid individuals in mosquitoes, as a result of the drug, is given by the constants $W_{RR}$, $W_{RS}$, and $W_{SS}$. What are the frequencies of the three different types after this differential survival?

**3.** After differential survival the three types produce several haploid individuals that infect humans. Suppose that each diploid individual produces a total of $b$ haploid




1. Infected mosquito bites a human. Sporozoites enter blood, which carries them to the liver.
2. Sporozoites reproduce asexually in liver cells, mature into merozoites. Merozoites leave the liver and infect red blood cells.
3. Merozoites reproduce asexually in some red blood cells.
4. In other red blood cells, merozoites differentiate into gametocytes.
5. A female mosquito bites and sucks blood from the infected person. Gametocytes in blood enter her gut and mature into gametes, which fuse to form zygotes.
6. Meiosis of zygotes produces cells that develop into sporozoites. The sporozoites migrate to the mosquito's salivary glands.

**FIGURE 1**

Life cycle of malaria

*Source:* From Starr. *Biology,* 8E © 2011 Brooks/Cole, a part of Cengage Learning, Inc. Reproduced by permission. www.cengage.com/permissions

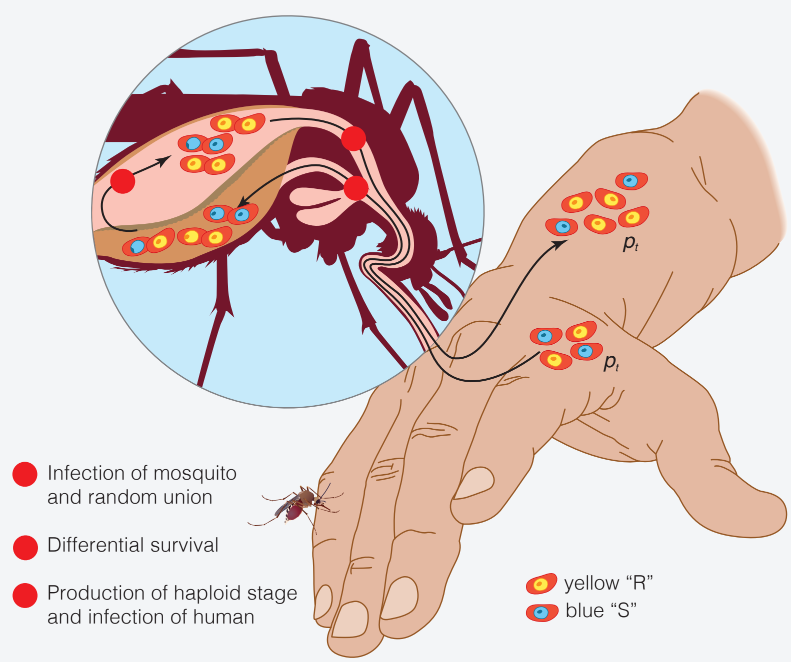



**FIGURE 2**

*Source:* Adapted from C. Starr et al., *Biology: Concepts and Applications*, 8th ed. (Belmont, CA: Cengage Learning, 2011), 316.

descendants, but RR individuals produce all R-type haploids, SS individuals produce all S-type haploids, and RS individuals produce a 50:50 mixture of both. What is the frequency of the R-type haploids in humans after this occurs?

We will assume the frequency doesn't change as the haploid parasites go through the rest of their life cycle in humans (as shown in Figure 1). Thus, the frequency just calculated is the value of $p$ after the completion of the entire life cycle and so it is equal to $p_{t+1}$. Therefore you should arrive at the final result

$$p_{t+1} = \frac{p_t^2 W_{\text{RR}} + p_t(1 - p_t)W_{\text{RS}}}{p_t^2 W_{\text{RR}} + 2p_t(1 - p_t)W_{\text{RS}} + (1 - p_t)^2 W_{\text{SS}}}$$

**4.** When the drug is removed from use we sometimes see that the frequency of the drug-resistant gene decreases because the S type survives best in the absence of the drug. For example, this might be modeled by choosing $W_{\text{RR}} = \frac{1}{4}$ and $W_{\text{RS}} = W_{\text{SS}} = \frac{1}{2}$. Show that, with these choices, the expression in Problem 3 reduces to the rational function discussed in Section 1.2 on page 25.

# Chapter 1 REVIEW

## CONCEPT CHECK

**1.** (a) What is a function? What are its domain and range?
(b) What is the graph of a function?
(c) How can you tell whether a given curve is the graph of a function?

**2.** Discuss four ways of representing a function. Illustrate your discussion with examples.

**3.** (a) What is an even function? How can you tell if a function is even by looking at its graph?
(b) What is an odd function? How can you tell if a function is odd by looking at its graph?

**4.** What is an increasing function?

**5.** What is a mathematical model?

**6.** Give an example of each type of function.
(a) Linear function (b) Power function
(c) Exponential function (d) Quadratic function
(e) Polynomial of degree 5 (f) Rational function

**7.** Sketch by hand, on the same axes, the graphs of the following functions.
(a) $f(x) = x$ (b) $g(x) = x^2$
(c) $h(x) = x^3$ (d) $j(x) = x^4$

**8.** Draw, by hand, a rough sketch of the graph of each function.
(a) $y = \sin x$ (b) $y = \cos x$
(c) $y = \tan x$ (d) $y = e^x$
(e) $y = \ln x$ (f) $y = 1/x$
(g) $y = |x|$ (h) $y = \sqrt{x}$

**9.** Suppose that $f$ has domain $A$ and $g$ has domain $B$.
(a) What is the domain of $f + g$?
(b) What is the domain of $fg$?
(c) What is the domain of $f/g$?

**10.** How is the composite function $f \circ g$ defined? What is its domain?

**11.** Suppose the graph of $f$ is given. Write an equation for each of the graphs that are obtained from the graph of $f$ as follows.
(a) Shift 2 units upward.
(b) Shift 2 units downward.
(c) Shift 2 units to the right.
(d) Shift 2 units to the left.
(e) Reflect about the $x$-axis.
(f) Reflect about the $y$-axis.
(g) Stretch vertically by a factor of 2.
(h) Shrink vertically by a factor of 2.
(i) Stretch horizontally by a factor of 2.
(j) Shrink horizontally by a factor of 2.

**12.** (a) What is a one-to-one function? How can you tell if a function is one-to-one by looking at its graph?
(b) If $f$ is a one-to-one function, how is its inverse function $f^{-1}$ defined? How do you obtain the graph of $f^{-1}$ from the graph of $f$?

**13.** (a) What is a semilog plot?
(b) If a semilog plot of your data lies approximately on a line, what type of model is appropriate?

**14.** (a) What is a log-log plot?
(b) If a log-log plot of your data lies approximately on a line, what type of model is appropriate?

**15.** (a) What is a sequence?
(b) What is a recursive sequence?

**16. Discrete-time models**
(a) If there are $N_t$ cells at time $t$ and they divide according to the difference equation $N_{t+1} = RN_t$, write an expression for $N_t$.
(b) If a population has carrying capacity $K$, write the logistic difference equation for $N_t$.
(c) Write the logistic difference equation for

$$x_t = \frac{r}{(1+r)K} N_t$$

**Answers to the Concept Check can be found on the back endpapers.**

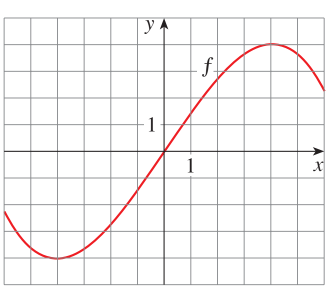
## TRUE-FALSE QUIZ

Determine whether the statement is true or false. If it is true, explain why. If it is false, explain why or give an example that disproves the statement.

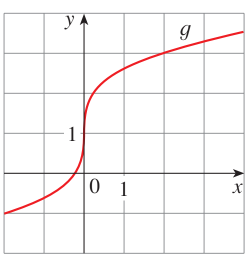
1. If $f$ is a function, then $f(s + t) = f(s) + f(t)$.
2. If $f(s) = f(t)$, then $s = t$.
3. If $f$ is a function, then $f(3x) = 3f(x)$.
4. If $x_1 < x_2$ and $f$ is a decreasing function, then $f(x_1) > f(x_2)$.
5. A vertical line intersects the graph of a function at most once.
6. If $f$ and $g$ are functions, then $f \circ g = g \circ f$.
7. If $f$ is one-to-one, then $f^{-1}(x) = \dfrac{1}{f(x)}$.
8. You can always divide by $e^x$.
9. If $0 < a < b$, then $\ln a < \ln b$.
10. If $x > 0$, then $(\ln x)^6 = 6 \ln x$.
11. If $x > 0$ and $a > 1$, then $\dfrac{\ln x}{\ln a} = \ln \dfrac{x}{a}$.
12. If $x$ is any real number, then $\sqrt{x^2} = x$.

## EXERCISES

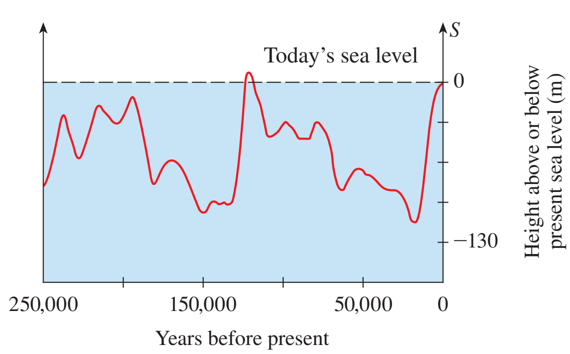
**1.** Let $f$ be the function whose graph is given.
(a) Estimate the value of $f(2)$.
(b) Estimate the values of $x$ such that $f(x) = 3$.
(c) State the domain of $f$.
(d) State the range of $f$.
(e) On what interval is $f$ increasing?
(f) Is $f$ one-to-one? Explain.
(g) Is $f$ even, odd, or neither even nor odd? Explain.




**2.** The graph of $g$ is given.
(a) State the value of $g(2)$.
(b) Why is $g$ one-to-one?
(c) Estimate the value of $g^{-1}(2)$.
(d) Estimate the domain of $g^{-1}$.
(e) Sketch the graph of $g^{-1}$.



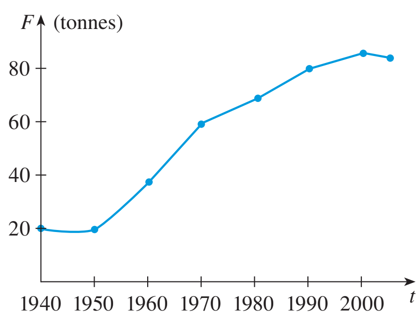

**3. Sea level** The figure shows how the sea level has changed over the last quarter of a million years according to data from




*Source:* Adapted from T. Garrison, *Oceanography: An Invitation to Marine Science* (Belmont, CA: Cengage Learning, 2010), 113, figure 4.15.

ocean floor cores. $S(t)$ is the sea level (in meters) relative to present sea level.
(a) What was the sea level 100,000 years ago?
(b) When was the sea level lowest? Highest?
(c) What is the range of this function?
(d) Can you account for the fluctuation of the sea level in terms of ice ages?

**4. Marine fish catch** The figure shows the worldwide commercial marine fish catch $F(t)$ in millions of tonnes (metric tons).
(a) In what year was the fish catch 70 million tonnes?
(b) What is the range of $F$?




*Source:* Adapted from T. Garrison, *Oceanography: An Invitation to Marine Science* (Belmont, CA: Cengage Learning, 2010), 472, figure 17.13a.

**5.** If $f(x) = x^2 - 2x + 3$, evaluate the difference quotient

$$\frac{f(a+h) - f(a)}{h}$$

**6.** Sketch a rough graph of the yield of a crop as a function of the amount of fertilizer used.

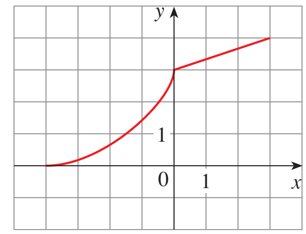
**7–10** Find the domain and range of the function. Write your answer in interval notation.

**7.** $f(x) = 2/(3x - 1)$ **8.** $g(x) = \sqrt{16 - x^4}$

**9.** $h(x) = \ln(x + 6)$ **10.** $F(t) = 3 + \cos 2t$

**11.** Suppose that the graph of $f$ is given. Describe how the graphs of the following functions can be obtained from the graph of $f$.
(a) $y = f(x) + 8$ (b) $y = f(x + 8)$
(c) $y = 1 + 2f(x)$ (d) $y = f(x - 2) - 2$
(e) $y = -f(x)$ (f) $y = f^{-1}(x)$

**12.** The graph of $f$ is given. Draw the graphs of the following functions.
(a) $y = f(x - 8)$ (b) $y = -f(x)$
(c) $y = 2 - f(x)$ (d) $y = \frac{1}{2}f(x) - 1$
(e) $y = f^{-1}(x)$ (f) $y = f^{-1}(x + 3)$




**13–19** Use transformations to sketch the graph of the function.

**13.** $y = -\sin 2x$ **14.** $y = 3\ln(x - 2)$

**15.** $y = \frac{1}{2}(1 + e^x)$ **16.** $y = 2 - \sqrt{x}$

**17.** $f(x) = \dfrac{1}{x + 2}$ **18.** $f(x) = x^2 - 2x$

**19.** $f(x) = \begin{cases} -x & \text{if } x < 0 \\ e^x - 1 & \text{if } x \geqslant 0 \end{cases}$

**20.** Determine whether $f$ is even, odd, or neither even nor odd.
(a) $f(x) = 2x^5 - 3x^2 + 2$
(b) $f(x) = x^3 - x^7$
(c) $f(x) = e^{-x^2}$
(d) $f(x) = 1 + \sin x$

**21.** If $f(x) = \ln x$ and $g(x) = x^2 - 9$, find the functions (a) $f \circ g$, (b) $g \circ f$, (c) $f \circ f$, (d) $g \circ g$, and their domains.

**22.** Express the function $F(x) = 1/\sqrt{x + \sqrt{x}}$ as a composition of three functions.

BB **23. Life expectancy** Life expectancy improved dramatically in the 20th century. The table gives the life expectancy at birth (in years) of males born in the United States. Use a scatter plot to choose an appropriate type of model. Use your model to predict the life-span of a male born in the year 2010.

| Birth year | Life expectancy | Birth year | Life expectancy |
|---|---|---|---|
| 1900 | 48.3 | 1960 | 66.6 |
| 1910 | 51.1 | 1970 | 67.1 |
| 1920 | 55.2 | 1980 | 70.0 |
| 1930 | 57.4 | 1990 | 71.8 |
| 1940 | 62.5 | 2000 | 73.0 |
| 1950 | 65.6 | | |

**24.** A small-appliance manufacturer finds that it costs \$9000 to produce 1000 toaster ovens a week and \$12,000 to produce 1500 toaster ovens a week.
(a) Express the cost as a function of the number of toaster ovens produced, assuming that it is linear. Then sketch the graph.

(b) What is the slope of the graph and what does it represent?
(c) What is the $y$-intercept of the graph and what does it represent?

**25.** If $f(x) = 2x + \ln x$, find $f^{-1}(2)$.

**26.** Find the inverse function of $f(x) = \dfrac{x+1}{2x+1}$.

**27.** Find the exact value of each expression.
(a) $e^{2\ln 3}$ (b) $\log_{10} 25 + \log_{10} 4$

**28.** Solve each equation for x.
(a) $e^x = 5$ (b) $\ln x = 2$
(c) $e^{e^x} = 2$

**29.** The half-life of palladium-100, $^{100}$Pd, is four days. (So half of any given quantity of $^{100}$Pd will disintegrate in four days.) The initial mass of a sample is one gram.
(a) Find the mass that remains after 16 days.
(b) Find the mass $m(t)$ that remains after $t$ days.
(c) Find the inverse of this function and explain its meaning.
(d) When will the mass be reduced to 0.01 g?

**30. Population growth** The population of a certain species in a limited environment with initial population 100 and carrying capacity 1000 is

$$P(t) = \frac{100{,}000}{100 + 900e^{-t}}$$

where $t$ is measured in years.
(a) Graph this function and estimate how long it takes for the population to reach 900.
(b) Find the inverse of this function and explain its meaning.
(c) Use the inverse function to find the time required for the population to reach 900. Compare with the result of part (a).

**31.** Graph members of the family of functions $f(x) = \ln(x^2 - c)$ for several values of $c$. How does the graph change when $c$ changes?

**32.** Graph the three functions $y = x^b$, $y = b^x$, and $y = \log_b x$ on the same screen for two or three values of $b > 1$. For large values of $x$, which of these functions has the largest values and which has the smallest values?

**33–34** Data points $(x, y)$ are given.
(a) Draw a scatter plot of the data points.
(b) Make semilog and log-log plots of the data.
(c) Is a linear, power, or exponential function appropriate for modeling these data?
(d) Find an appropriate model for the data and then graph the model together with a scatter plot of the data.

**33.**

| $x$ | 4 | 8 | 12 | 16 | 20 | 24 |
|---|---|---|---|---|---|---|
| $y$ | 7.0 | 11.5 | 15.2 | 18.9 | 22.1 | 25.0 |

**34.**

| $x$ | 1 | 3 | 6 | 10 | 14 | 16 |
|---|---|---|---|---|---|---|
| $y$ | 7.22 | 4.61 | 2.38 | 0.99 | 0.41 | 0.26 |

**35. Nigerian population** The table gives the midyear population of Nigeria (in millions) from 1985 to 2010.

| Year | Population | Year | Population |
|---|---|---|---|
| 1985 | 85 | 2000 | 124 |
| 1990 | 97 | 2005 | 142 |
| 1995 | 110 | 2010 | 162 |

(a) Make a scatter plot, semilog plot, and log-log plot for these data and comment on which type of model would be most appropriate.
(b) Obtain an exponential model for the population.
(c) Use your model to estimate the population in 2008 and predict the population in 2020.

**36–37** Find the first six terms of the sequence.

**36.** $a_n = \sin(n\pi/3)$

**37.** $a_1 = 3, \ a_{n+1} = n + 2a_n - 1$

**38.** Find a formula for the general term of the sequence

$$-3, \ \frac{5}{4}, \ -\frac{7}{9}, \ \frac{9}{16}, \ -\frac{11}{25}, \ \ldots$$

assuming that the pattern of the first few terms continues.

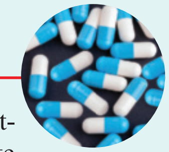
**39.** If $x_0 = 0.9$ and $x_{t+1} = 2.7x_t(1 - x_t)$, calculate $x_t$ to four decimal places for $t = 1, 2, \ldots 10$ and graph $x_t$. Comment on the behavior of the sequence.

**40. Beverton-Holt model** An alternative to the logistic model for restricted population growth is the *Beverton-Holt recruitment curve*. Here the recursion model is

$$N_{t+1} = \frac{cN_t}{1 + (c-1)N_t/K}$$

where $K$ is the carrying capacity and $c$ is the per capita growth factor.
(a) If $K = 50$ and $c = 1.7$, plot some points on the graph of $N_t$ for the following values of the initial population: $N_0 = 10, 30, 70$.
(b) For the values in part (a), compare the Beverton-Holt model with the logistic model.

## CASE STUDY 1a Kill Curves and Antibiotic Effectiveness



We are studying the relationship between the magnitude of antibiotic treatment and the effectiveness of the treatment. Recall that the extent of bacterial killing by an antibiotic is determined by both the *antibiotic concentration profile* and the *dose response relationship*. Figures 1 and 2 show these plots for the antibiotic ciprofloxacin when used against *E. coli*.[1]

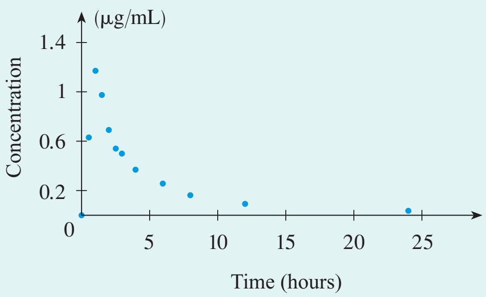



**FIGURE 1**
Antibiotic concentration profile in plasma of a healthy human volunteer after receiving 500 mg of ciprofloxacin

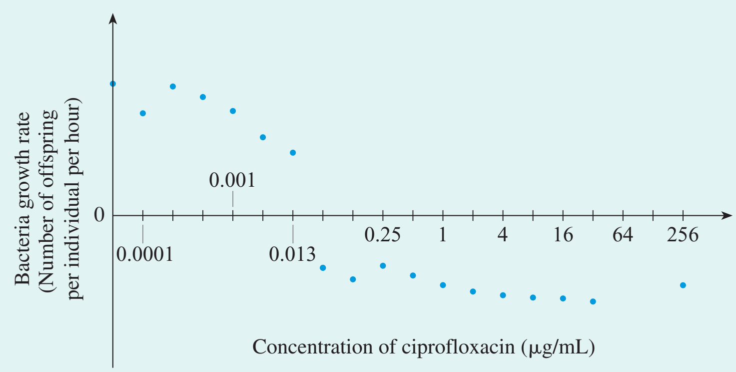



**FIGURE 2**
Dose response relationship for ciprofloxacin with the bacteria *E. coli*

Now, in the words of Picasso, we are viewing mathematical models as "lies that reveal truth." In other words, we don't expect our mathematical model to capture every detail of the biological system; rather, we simply want it to capture the most important features. To this end, let's describe the main patterns seen in Figures 1 and 2 mathematically.

Figure 1 shows that the antibiotic concentration increases extremely quickly, followed by a slow decay. To simplify matters let's therefore suppose that it increases instantly from zero to the peak concentration at time $t = 0$, and it then decays. As we will see in Case Study 1b, the decay can be well modeled using the exponential decay function

**(1)** $$c(t) = c_0 e^{-kt}$$

where $c_0$ is the concentration at $t = 0$ and $k$ is a positive constant. Equation 1 is plotted in Figure 3.

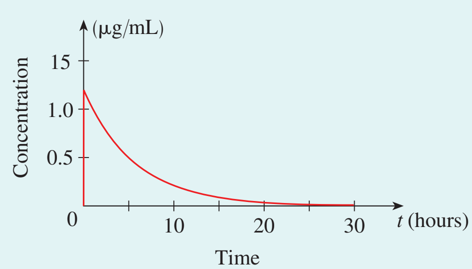



**FIGURE 3**
Drug concentration profile modeled by the function $c(t) = c_0 e^{-kt}$ with $c_0 = 1.2$ μg/mL and $k = 0.175$

1. Adapted from S. Imre et al., "Validation of an HPLC Method for the Determination of Ciprofloxacin in Human Plasma," *Journal of Pharmaceutical and Biomedical Analysis* 33 (2003): 125–30.

In Figure 2 it looks like the dose response relationship doesn't vary much up to a concentration of around 0.013 μg/mL. It then drops suddenly to a low value and remains relatively constant as the antibiotic concentration increases further. To simplify matters, let's model the dose response relationship by the piecewise defined function

$$r(c) = \begin{cases} r_2 & \text{if } c < MIC \\ r_1 & \text{if } c \geqslant MIC \end{cases}$$

where $MIC$ is a constant that is referred to as the *minimum inhibitory concentration* ($MIC = 0.013$ μg/mL in this case), $r_1$ and $r_2$ are constants giving the bacteria population growth rate under high and low antibiotic concentration, respectively, and $r_1 < 0$ and $r_2 > 0$. This function is plotted in Figure 4.[2]

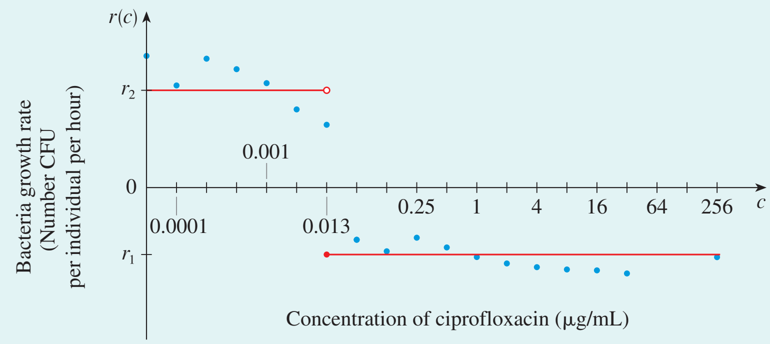



**FIGURE 4**
Dose response relationship modeled by the piecewise defined function $r(c)$

The functions in Figures 3 and 4 will, together, determine how the bacteria population size changes over time. At $t = 0$ the antibiotic concentration is $c_0$, and if $c_0$ is greater than $MIC = 0.013$ μg/mL, then the bacteria population size will decline. At the same time the antibiotic concentration will decay as time passes, eventually reaching a value of $MIC = 0.013$ μg/mL. At this point the growth rate of the bacteria population becomes positive.

In Case Study 1b you will show that, using the functions in Figures 3 and 4, a suitable model for the size of the bacteria population $P(t)$ (in CFU/mL) as a function of time $t$ (in hours) is given by the piecewise defined function

**(2a)**
$$P(t) = \begin{cases} 6e^{t/3} & \text{if } t < 2.08 \\ 12 & \text{if } t \geqslant 2.08 \end{cases}$$

if $c_0 < 0.013$, and

**(2b)**
$$P(t) = \begin{cases} 6e^{-t/20} & \text{if } t < a \\ 6Ae^{t/3} & \text{if } a \leqslant t < b \\ 12 & \text{if } t \geqslant b \end{cases}$$

if $c_0 \geqslant 0.013$, where the constants $a$, $b$, and $A$ are defined by $a = 5.7 \ln(77c_0)$, $b = 6.6 \ln(77c_0) + 2.08$, and $A = (77c_0)^{-2.2}$.

2. Adapted from W. Bär et al., "Rapid Method for Detection of Minimal Bactericidal Concentration of Antibiotics," *Journal of Microbiological Methods* 77 (2009): 85–89, Figure 1.

**1.** Plot $P(t)$ as a function of time for each of the concentrations $c_0 = 0, 0.019, 0.038, 0.075, 0.15, 0.3, 0.6, 1.2$. These are the kill curves predicted by the model. Comment on the similarities and differences between the predicted curves and those from the data in Figure 5.[3]

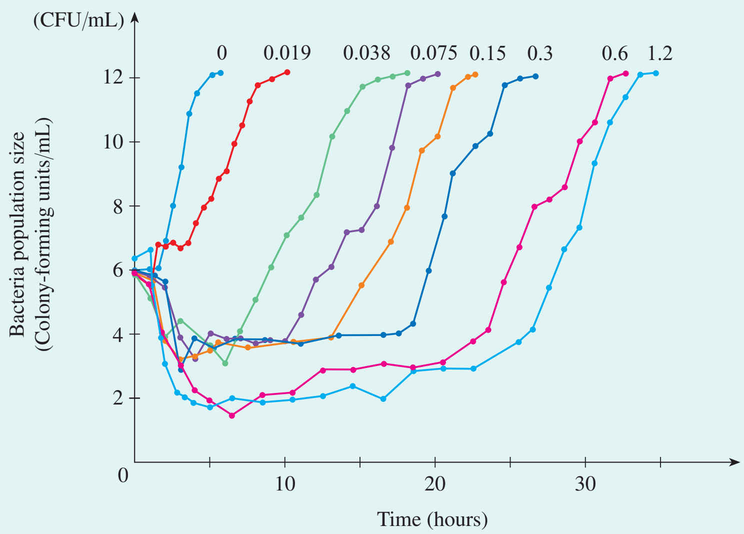



**FIGURE 5**
The kill curves of ciprofloxacin for *E. coli* when measured in a growth chamber. The concentration of ciprofloxacin at $t = 0$ is indicated above each curve (in $\mu$g/mL).

Our goal is to summarize the model kill curves from Problem 1 in a simpler form in order to see more clearly the relationship between the magnitude of antibiotic treatment and its predicted effectiveness.

To do this, we need to obtain a measure of the magnitude of antibiotic treatment as well as a measure of its effectiveness. We first obtain a measure of the magnitude of antibiotic treatment from the antibiotic concentration profiles that underlie each predicted kill curve. Three measures are commonly used: (1) the peak antibiotic concentration divided by *MIC*, denoted by $\rho$; (2) the duration of time for which the antibiotic concentration remains above *MIC*, denoted by $\tau$; and (3) the area under the antibiotic concentration profile divided by *MIC*, denoted by $\alpha$. These measures are illustrated graphically in Figure 6.

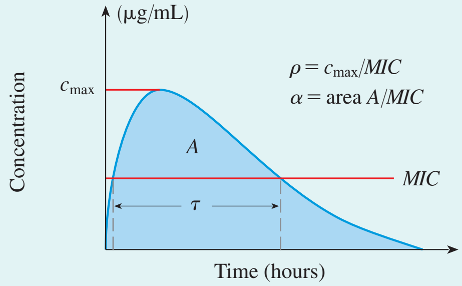



**FIGURE 6**
Three measures $\rho$, $\alpha$, and $\tau$ of the magnitude of antibiotic treatment

3. Adapted from A. Firsov et al., "Parameters of Bacterial Killing and Regrowth Kinetics and Antimicrobial Effect Examined in Terms of Area under the Concentration-Time Curve Relationships: Action of Ciprofloxacin against *Escherichia coli* in an In Vitro Dynamic Model." *Antimicrobial Agents and Chemotherapy* 41 (1997): 1281–87.

**2.** Find expressions for $\rho$ and $\tau$ in terms of $k$, $c_0$, and $MIC$, using Equation 1 for the antibiotic concentration profile.

**3.** In Case Study 1c you will show that $\alpha = \dfrac{1}{k}\dfrac{c_0}{MIC}$. Sketch graphs of $\rho$, $\tau$, and $\alpha$ as functions of $c_0$, using the values $k = 0.175$ (1/hours) and $MIC = 0.013$ µg/mL. What are their units?

You will notice from Problem 3 that, for a given antibiotic and bacterial species (in other words, for a given value of $k$ and $MIC$), all three quantities $\rho$, $\tau$, and $\alpha$ increase with one another. For example, it is not possible to have a high value of $\alpha$ without also having high values of $\rho$ and $\tau$. Therefore, since these measures are not independent of one another, we need to consider only one of them. We will focus the remainder of our study on $\alpha$, since it is the most commonly used.

Next we need to quantify the effectiveness of the antibiotic by quantifying different properties of the kill curves. Let's consider two possibilities: (i) the time taken to reduce the bacteria population size to 90% of its initial size, denoted by $T$, and (ii) the drop in population size before the population rebound occurs, denoted by $\Delta$. Both measures are shown in Figure 7.

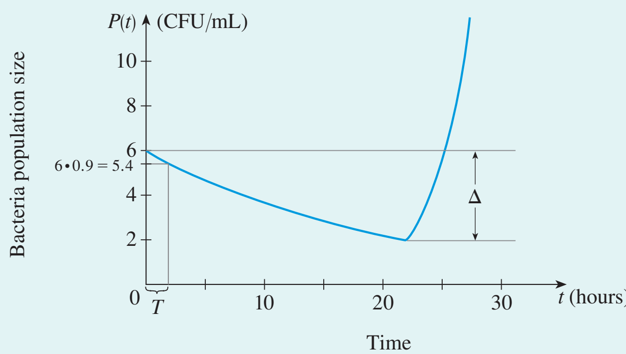



**FIGURE 7**
Two measures $T$ and $\Delta$ of antibiotic effectiveness

**4.** Find expressions for $\Delta$ and $T$ in the modeled populations in terms of $c_0$.

Our final goal is to use the results from Problem 4, along with the expression for $\alpha$, to plot $\Delta$ against $\alpha$ and to plot $T$ against $\alpha$ as well. This will give us the model's predictions for the plots in Figures 5 and 6 in Case Study 1 on page xlii.

**5.** Substitute the values $k = 0.175$ and $MIC = 0.013$ into the expression for $\alpha$. This expression, along with the results from Problem 4, should give you functions of the form $T = f(c_0)$, $\Delta = g(c_0)$, and $\alpha = h(c_0)$ for some functions $f$, $g$, and $h$. [*Note:* Some of the functions might actually be independent of $c_0$.]

**6.** Using the concept of inverse functions, explain how to obtain a function that gives $\Delta$ as a function of $\alpha$ in terms of $g$ and $h^{-1}$. Find an explicit expression for this function.

**7.** What is $T$ as a function of $\alpha$?

**8.** Plot the functions obtained in Problems 6 and 7. You should obtain the curves shown in Figures 8 and 9.

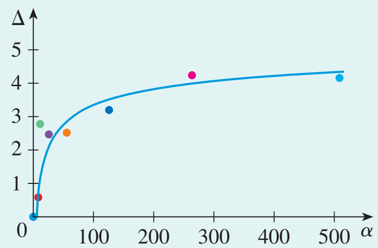



**FIGURE 8**
Predicted relationship between $\Delta$ and $\alpha$, along with the observations obtained using the kill curve data in Figure 5

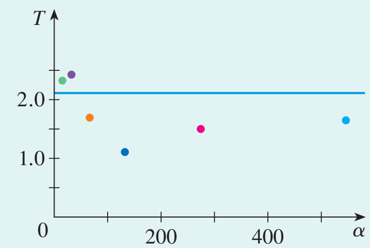
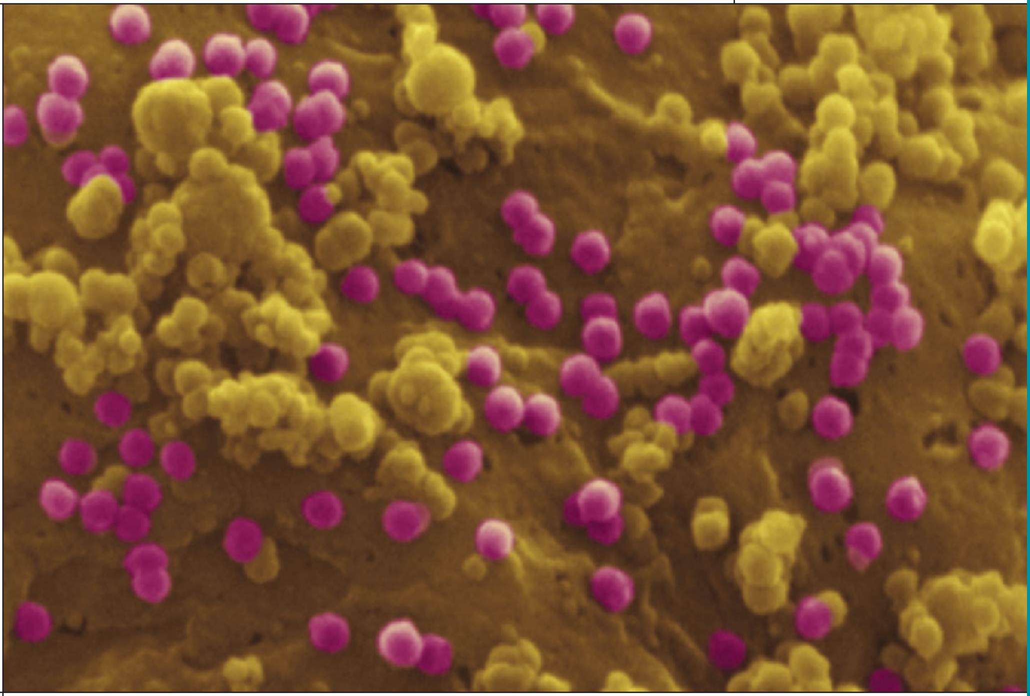



**FIGURE 9**
Predicted relationship between $T$ and $\alpha$, along with the observations obtained using the kill curve data in Figure 5

**9.** From Figures 8 and 9 you can see that this relatively simple model predicts the observed data reasonably well. In particular, $T$ is predicted to be independent of the magnitude of antibiotic treatment, whereas $\Delta$ increases with it. Provide a biological explanation in terms of the model for why this occurs. [*Hint:* Relate the fact that $T$ is predicted to be independent of $\alpha$ to the form of the kill curves from Problem 1 for different antibiotic doses.]

# Limits

# 2



Viruses infect all living organisms and are responsible for diseases ranging from the common cold to smallpox and AIDS. In the project on page 101 you are asked to use recursive sequences to investigate the interaction among viral infection, the human immune system, and antiviral drugs.

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**THE IDEA OF A LIMIT** is the basic concept in all of calculus. It underlies such phenomena as the long-term behavior of a population, the rate of growth of a tumor, and the area of a leaf.

# 2.1 Limits of Sequences

## The Long-Term Behavior of a Sequence

In Section 1.6 we looked at sequences, both those given by a simple defining formula and those defined recursively by a difference equation. Here we investigate what happens to the terms $a_n$ of a sequence in the long run. In other words, we explore what happens to $a_n$ as $n$ becomes large.

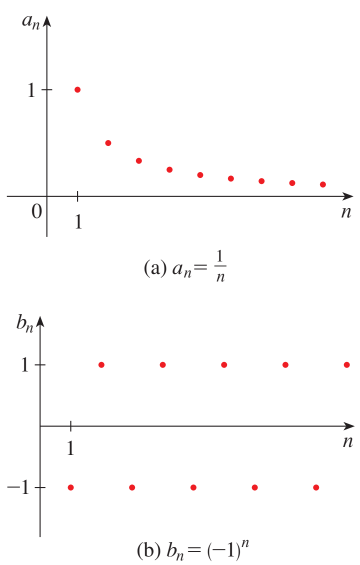



**FIGURE 1**

**EXAMPLE 1** | What happens to the terms of the sequence when $n$ becomes large?

(a) $a_n = \dfrac{1}{n}$ (b) $b_n = (-1)^n$

**SOLUTION**

(a) The first few terms of the sequence are

$$1, \quad \frac{1}{2}, \quad \frac{1}{3}, \quad \frac{1}{4}, \quad \frac{1}{5}, \quad \frac{1}{6}, \quad \ldots$$

and for larger values of $n$ we have

$$a_{10} = 0.1, \quad a_{100} = 0.01, \quad a_{1000} = 0.001, \quad a_{1,000,000} = 0.000001, \quad \ldots$$

The larger the value of $n$, the smaller the value of $a_n$. The terms are approaching 0 as $n$ increases. [See Figure 1(a).]

(b) Here the terms are

$$-1, \quad 1, \quad -1, \quad 1, \quad -1, \quad 1, \quad -1, \quad \ldots$$

The values of the terms alternate between 1 and $-1$ forever. So they don't approach any fixed number. [See Figure 1(b)]. ■

The sequences in Example 1 behave quite differently. The terms $a_n = 1/n$ approach 0 as $n$ becomes large. (In fact we could make $1/n$ as small as we like by taking $n$ large enough). We indicate this by saying that the sequence has *limit* 0 and by writing

$$\lim_{n \to \infty} \frac{1}{n} = 0$$

On the other hand, the sequence $b_n = (-1)^n$ does not have a limit, that is,

$$\lim_{n \to \infty} (-1)^n \quad \text{does not exist}$$

## Definition of a Limit

In general we write

$$\lim_{n \to \infty} a_n = L$$

if the terms $a_n$ approach $L$ as $n$ becomes large.

A more precise definition of the limit of a sequence is given in Appendix D.

**(1) Definition** A sequence $\{a_n\}$ has the **limit** $L$ and we write

$$\lim_{n\to\infty} a_n = L \qquad \text{or} \qquad a_n \to L \text{ as } n \to \infty$$

if we can make the terms $a_n$ as close to $L$ as we like by taking $n$ sufficiently large. If $\lim_{n\to\infty} a_n$ exists, we say the sequence **converges** (or is **convergent**). Otherwise, we say the sequence **diverges** (or is **divergent**).

Figure 2 illustrates Definition 1 by showing the graphs of two sequences that have the limit $L$.

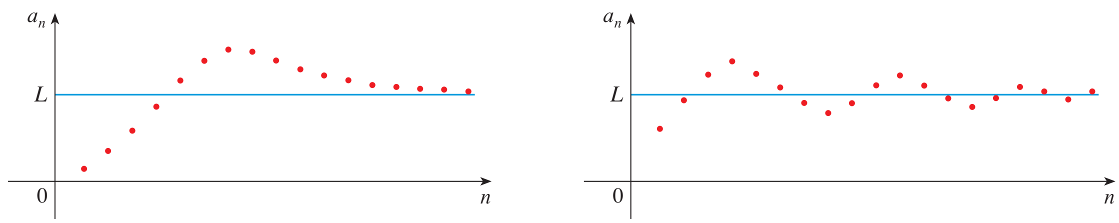



**FIGURE 2**
Graphs of two sequences with $\lim_{n\to\infty} a_n = L$

If $a_n$ becomes large as $n$ becomes large, we use the notation

$$\lim_{n\to\infty} a_n = \infty$$

In this case the sequence $\{a_n\}$ is divergent, but in a special way. We say that $a_n$ diverges to $\infty$.

**EXAMPLE 2** | Is the sequence $a_n = \sqrt{n}$ convergent or divergent?

**SOLUTION** When $n$ is large, $\sqrt{n}$ is large, so

$$\lim_{n\to\infty} \sqrt{n} = \infty$$

and the sequence $\{\sqrt{n}\}$ is divergent. Notice that we can make $\sqrt{n}$ as big as we want by taking $n$ big enough. For instance,

$$\sqrt{n} > 1000 \qquad \text{when} \qquad n > 1{,}000{,}000$$ ■

## ■ Limit Laws

The more precise version of Definition 1 in Appendix D can be used to prove the following properties of limits.

Limit Laws for Sequences

If $\{a_n\}$ and $\{b_n\}$ are convergent sequences and $c$ is a constant, then

$$\lim_{n\to\infty} (a_n + b_n) = \lim_{n\to\infty} a_n + \lim_{n\to\infty} b_n$$

$$\lim_{n\to\infty} (a_n - b_n) = \lim_{n\to\infty} a_n - \lim_{n\to\infty} b_n$$

$$\lim_{n\to\infty} ca_n = c \lim_{n\to\infty} a_n \qquad\qquad \lim_{n\to\infty} c = c$$

$$\lim_{n\to\infty} (a_n b_n) = \lim_{n\to\infty} a_n \cdot \lim_{n\to\infty} b_n$$

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{\lim_{n\to\infty} a_n}{\lim_{n\to\infty} b_n} \quad \text{if } \lim_{n\to\infty} b_n \neq 0$$

$$\lim_{n\to\infty} a_n^p = \left[\lim_{n\to\infty} a_n\right]^p \quad \text{if } p > 0 \text{ and } a_n > 0$$

From the last of these laws and the fact that $\lim_{n\to\infty} (1/n) = 0$, we deduce that

**(2)** $$\lim_{n\to\infty} \frac{1}{n^p} = 0 \qquad \text{for any number } p > 0$$

Combining this fact with the various limit laws, we can calculate limits of sequences as in the following example.

**EXAMPLE 3** | Find $\displaystyle\lim_{n\to\infty} \frac{1 + 2n^2}{5 + 3n + 4n^2}$.

**SOLUTION** As $n$ becomes large, both numerator and denominator become large, so it isn't obvious what happens to their ratio. If we divide the numerator and denominator by $n^2$ (the highest power of $n$ that occurs in the denominator), then we can use the Limit Laws and take advantage of the limits we know from Equation 2:

$$\lim_{n\to\infty} \frac{1 + 2n^2}{5 + 3n + 4n^2} = \lim_{n\to\infty} \frac{\dfrac{1 + 2n^2}{n^2}}{\dfrac{5 + 3n + 4n^2}{n^2}} = \lim_{n\to\infty} \frac{\dfrac{1}{n^2} + 2}{\dfrac{5}{n^2} + \dfrac{3}{n} + 4}$$

$$= \frac{\displaystyle\lim_{n\to\infty} \frac{1}{n^2} + \lim_{n\to\infty} 2}{\displaystyle 5\lim_{n\to\infty} \frac{1}{n^2} + 3\lim_{n\to\infty} \frac{1}{n} + \lim_{n\to\infty} 4}$$

$$= \frac{0 + 2}{5 \cdot 0 + 3 \cdot 0 + 4} = \frac{1}{2}$$

We have used the Limit Laws and Equation 2 with $p = 2$ and $p = 1$. ■

## ■ Geometric Sequences

A **geometric sequence** is a sequence of the form $b_n = ar^n$. We start with a number $b_0 = a$ and multiply repeatedly by a number $r$:

$$a, \; ar, \; ar^2, \; ar^3, \; ar^4, \; \ldots$$

We saw an example of this in Section 1.4 with $a = 1$ and $r = 8$. The malarial species *P. chabaudi* reproduces every 24 hours and so a single such parasite at time $t = 0$ results in $b_n = 8^n$ parasites after $n$ days (at least initially). From our knowledge of the exponential function, we know that the sequence $b_n = 8^n$ grows indefinitely:

$$\lim_{n\to\infty} 8^n = \infty$$

We also modeled cell division by the recursion

$$N_{t+1} = RN_t$$

whose solution is

$$N_t = N_0 R^t$$

where $R$ is the per capita growth factor. So this is a geometric sequence with $a = N_0$ and $r = R$.

In general there are three cases for the geometric sequence $b_n = r^n$ if $r$ is a positive number. From the graphs of the exponential functions in Figures 1.4.4 and 1.4.5 (page 44) we see that

**(3)**
$$\boxed{\lim_{n\to\infty} r^n = \begin{cases} 0 & \text{if } 0 < r < 1 \\ 1 & \text{if } r = 1 \\ \infty & \text{if } r > 1 \end{cases}}$$

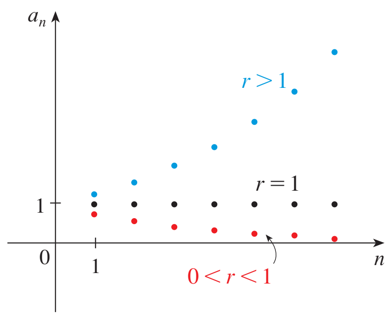



**FIGURE 3**
The sequence $b_n = r^n$

These three cases are illustrated in Figure 3.

**EXAMPLE 4** | Calculate $\displaystyle\lim_{n\to\infty} \frac{2^n - 1}{6^n}$ if it exists.

**SOLUTION** Simplifying, we get

$$\lim_{n\to\infty} \frac{2^n - 1}{6^n} = \lim_{n\to\infty} \left[\frac{2^n}{6^n} - \frac{1}{6^n}\right] = \lim_{n\to\infty} \left[\left(\frac{1}{3}\right)^n - \left(\frac{1}{6}\right)^n\right]$$

$$= \lim_{n\to\infty} \left(\frac{1}{3}\right)^n - \lim_{n\to\infty} \left(\frac{1}{6}\right)^n = 0 - 0 = 0$$

where the second last equality follows from (3) with $r = \frac{1}{3}$ and $r = \frac{1}{6}$. ■

## ■ Recursion for Medication

Let's modify the geometric recursion $b_{n+1} = rb_n$ by adding a constant term $c$:

$$b_{n+1} = rb_n + c$$

This difference equation arises when a patient is given a daily medication.

**EXAMPLE 5** | **BB** Drug concentration A drug is administered to a patient at the same time every day. Suppose the concentration of the drug is $C_n$ (measured in mg/mL) after the injection on the $n$th day. Before the injection the next day, only 30% of the drug present on the preceding day remains in the bloodstream. If the daily dose raises the concentration by 0.2 mg/mL, find the concentration after four days.

**SOLUTION** Just before the daily dose of medication is administered, the concentration is reduced to 30% of the preceding day's concentration, that is, $0.3C_n$. With the new dose, the concentration is increased by 0.2 mg/mL and so

**(4)**
$$\boxed{C_{n+1} = 0.3C_n + 0.2}$$

Starting with $C_0 = 0$ and putting $n = 0, 1, 2, 3$ into this difference equation, we get

$$C_1 = 0.3C_0 + 0.2 = 0.2$$

$$C_2 = 0.3C_1 + 0.2 = 0.3(0.2) + 0.2 = 0.26$$

$$C_3 = 0.3C_2 + 0.2 = 0.3(0.26) + 0.2 = 0.278$$

$$C_4 = 0.3C_3 + 0.2 = 0.3(0.278) + 0.2 = 0.2834$$

The concentration after four days is 0.2834 mg/mL. ■

The successive daily concentrations in Example 5 appear to be increasing. Do you think they increase indefinitely? Or do they have a limit?

For now, let's *assume* there is a limiting concentration. Let's call it $C$, that is, $\lim_{n\to\infty} C_n = C$. As $n \to \infty$, observe that $n + 1 \to \infty$ too. So $\lim_{n\to\infty} C_{n+1} = C$ too. Therefore, taking limits in Equation 4, we have

$$\begin{aligned}
\lim_{n\to\infty} C_{n+1} &= \lim_{n\to\infty} (0.3C_n + 0.2) = 0.3 \lim_{n\to\infty} C_n + \lim_{n\to\infty} 0.2 \\
C &= 0.3C + 0.2 \\
0.7C &= 0.2 \\
C &= \frac{0.2}{0.7} = \frac{2}{7}
\end{aligned}$$

This shows that *if* there is a limiting concentration $C$, then

$$C = \frac{2}{7} \approx 0.2857 \text{ mg/mL}$$

We are going to verify that there is indeed a limiting concentration by first finding an explicit formula for $C_n$.

## ■ Geometric Series

If we add the terms of a geometric sequence, we get another sequence, this one consisting of sums:

$$\begin{aligned}
s_0 &= a \\
s_1 &= a + ar \\
s_2 &= a + ar + ar^2 \\
s_3 &= a + ar + ar^2 + ar^3 \\
&\vdots \\
s_n &= a + ar + ar^2 + \cdots + ar^n
\end{aligned}$$

If we multiply $s_n$ by $r$ and align the terms with those above, we get

$$rs_n = \qquad ar + ar^2 + \cdots + ar^n + ar^{n+1}$$

If we now subtract these last two equations, most of the terms cancel in pairs:

$$s_n - rs_n = a - ar^{n+1}$$
$$s_n(1 - r) = a(1 - r^{n+1})$$

If $r \neq 1$, we can solve for $s_n$:

The sum of terms of a sequence is called a **series**. Formula 5 gives the sum of a finite geometric series.

**(5)**
$$\boxed{s_n = a + ar + ar^2 + \cdots + ar^n = \frac{a(1 - r^{n+1})}{1 - r}}$$

We know that the solution of the difference equation

$$b_{n+1} = rb_n \qquad b_0 = a$$

is $b_n = ar^n$. We can now use Formula 5 to find a solution of the recursion

$$b_{n+1} = c + rb_n \qquad b_0 = a$$

To see the pattern, we start by computing the first few terms:

$$\begin{aligned}
b_1 &= c + ra \\
b_2 &= c + rb_1 = c + r(c + ra) = c(1 + r) + r^2a \\
b_3 &= c + rb_2 = c + r(c + cr + r^2a) = c(1 + r + r^2) + r^3a \\
&\ \vdots \\
b_n &= c(1 + r + r^2 + \cdots r^{n-1}) + r^na
\end{aligned}$$

Using Formula 5 for the sum of a finite geometric series, we have the following formula.

**(6)** The solution of the difference equation $b_{n+1} = c + rb_n$, $b_0 = a$, is

$$b_n = r^na + c\left(\frac{1 - r^n}{1 - r}\right)$$

**EXAMPLE 6** | **Drug concentration (continued)** What is the concentration of the drug in Example 5 after the $n$th dose? What is the limiting concentration?

**SOLUTION** The difference equation in Example 5 is of the form $C_{n+1} = rC_n + c$, where $r = 0.3$, $c = 0.2$, and $C_0 = 0$. So from Equation 6 we have

$$C_n = 0.2\left[\frac{1 - (0.3)^n}{1 - 0.3}\right] = \frac{2}{7}[1 - (0.3)^n]$$

Because $0.3 < 1$, we know from Equation 3 that $\lim_{n\to\infty}(0.3)^n = 0$. So the limiting concentration is

$$\lim_{n\to\infty} C_n = \lim_{n\to\infty} \frac{2}{7}[1 - (0.3)^n] = \frac{2}{7}(1 - 0) = \frac{2}{7}\text{ mg/mL}$$

as we had predicted. ■

What happens if we try to add all of the infinitely many terms of a geometric sequence? For instance, if $a = 1$ and $r = \frac{1}{2}$, does it make sense to talk about the value of the infinite sum

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots + \frac{1}{2^n} + \cdots$$

This sum is called an **infinite series**. From Equation 5 we know that the sum of the first $n + 1$ terms is

$$s_n = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots + \frac{1}{2^n} = \frac{1\left(1 - \dfrac{1}{2^{n+1}}\right)}{1 - \frac{1}{2}} = 2\left(1 - \frac{1}{2^{n+1}}\right)$$

and $\lim_{n\to\infty}(1/2^{n+1}) = 0$, so $\lim_{n\to\infty} s_n = 2$. In other words, by adding enough terms of the series, we can make the sum as close as we like to 2. We say that the sum of the geometric series is 2 and we write

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots + \frac{1}{2^{n-1}} + \frac{1}{2^n} + \cdots = 2$$

In general, if $-1 < r < 1$, we know that $\lim_{n\to\infty} r^n = 0$ and so, from Equation 5,

$$\lim_{n\to\infty} s_n = \lim_{n\to\infty}\left[\frac{a}{1-r} - \frac{a}{1-r}r^{n+1}\right] = \frac{a}{1-r} - \frac{a}{1-r}\lim_{n\to\infty} r^{n+1} = \frac{a}{1-r}$$

If $-1 < r < 1$, the sum of the infinite geometric series is

$$a + ar + ar^2 + \cdots + ar^n + \cdots = \frac{a}{1-r}$$

As the following example illustrates, a repeating decimal number is the sum of an infinite geometric series.

**EXAMPLE 7** | Write the number $2.3\overline{17} = 2.3171717\ldots$ as a ratio of integers.

**SOLUTION**

$$2.3171717\ldots = 2.3 + \frac{17}{10^3} + \frac{17}{10^5} + \frac{17}{10^7} + \cdots$$

After the first term we have a geometric series with $a = 17/10^3$ and $r = 1/10^2$. Therefore

$$2.3\overline{17} = 2.3 + \frac{\dfrac{17}{10^3}}{1 - \dfrac{1}{10^2}} = 2.3 + \frac{\dfrac{17}{1000}}{\dfrac{99}{100}}$$

$$= \frac{23}{10} + \frac{17}{990} = \frac{1147}{495}$$

## The Logistic Sequence in the Long Run

In Section 1.6 we looked at the logistic difference equation, which is of the form

$$x_{t+1} = cx_t(1 - x_t) \tag{7}$$

where $c$ is a constant. Here we examine this equation again, this time exploring the limiting behavior of the terms for various values of $c$.

**EXAMPLE 8 | BB Logistic difference equation** Compute and plot the next 16 terms of the logistic equation (7) in the following cases. Then describe the limiting behavior of the sequence.

(a) $x_0 = 0.8,\ \ c = 2.8$ (b) $x_0 = 0.4,\ \ c = 3.4$

**SOLUTION**

(a) Taking $x_0 = 0.8$ and $x_{t+1} = 2.8x_t(1 - x_t)$, we use a graphing calculator or computer to compute the terms up to $x_{16}$ to four decimal places and then we plot the graph in Figure 4.

| $t$ | $x_t$ | $t$ | $x_t$ |
|---|---|---|---|
| 0 | 0.8000 | 9 | 0.6321 |
| 1 | 0.4480 | 10 | 0.6511 |
| 2 | 0.6924 | 11 | 0.6360 |
| 3 | 0.5963 | 12 | 0.6482 |
| 4 | 0.6740 | 13 | 0.6385 |
| 5 | 0.6152 | 14 | 0.6463 |
| 6 | 0.6628 | 15 | 0.6401 |
| 7 | 0.6258 | 16 | 0.6450 |
| 8 | 0.6557 | | |

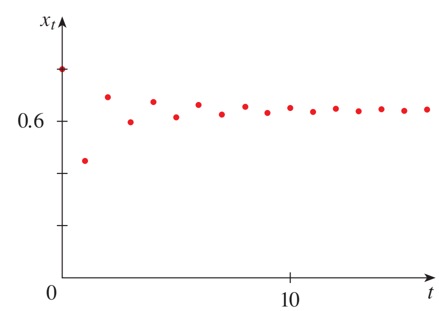



**FIGURE 4**

From the table of values and Figure 4 it appears that the terms of the sequence are approaching a number between 0.64 and 0.65. In fact if we assume that the limit exists and we call it $L$, we can evaluate it by the same technique as for the sequence in Example 5. We have both $\lim_{t\to\infty} x_t = L$ and $\lim_{t\to\infty} x_{t+1} = L$ and so, using the Limit Laws, we have

$$L = \lim_{t\to\infty} x_{t+1} = \lim_{t\to\infty} [2.8x_t(1 - x_t)] = 2.8L(1 - L)$$

$$1 = 2.8(1 - L)$$

$$1 - L = \frac{1}{2.8} \quad \Rightarrow \quad L = 1 - \frac{1}{2.8} = \frac{1.8}{2.8} = \frac{9}{14}$$

$$\lim_{n\to\infty} x_t = \frac{9}{14} \approx 0.64286$$

(b) Here $x_0 = 0.4$ and $x_{t+1} = 3.4x_t(1 - x_t)$. Again we calculate the next 16 terms and plot them in Figure 5.

| $t$ | $x_t$ | $t$ | $x_t$ |
|---|---|---|---|
| 0 | 0.4000 | 9 | 0.8385 |
| 1 | 0.8160 | 10 | 0.4603 |
| 2 | 0.5105 | 11 | 0.8446 |
| 3 | 0.8496 | 12 | 0.4461 |
| 4 | 0.4344 | 13 | 0.8401 |
| 5 | 0.8354 | 14 | 0.4566 |
| 6 | 0.4676 | 15 | 0.8436 |
| 7 | 0.8464 | 16 | 0.4486 |
| 8 | 0.4419 | | |

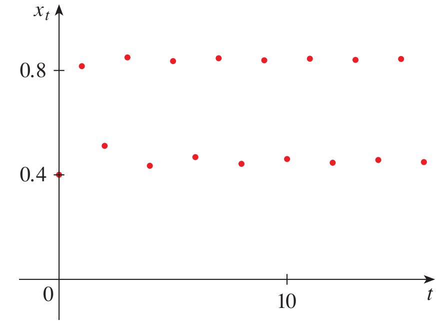



**FIGURE 5**

Looking at the table and Figure 5, it appears that the values of $x_t$ do not approach any fixed number. Instead, they oscillate between values near 0.84 and values near 0.45. In other words, for $c = 3.4$,

$$\lim_{t \to \infty} x_t \quad \text{does not exist}$$ ■

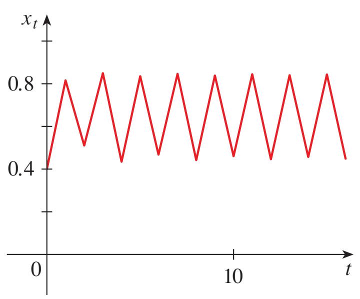



**FIGURE 6**

Sometimes it's easier to see what's happening in the graph of a sequence if we join consecutive points by line segments. In Figure 6 we have done this for the logistic sequence in Example 8(b) by joining the points in Figure 5. Thus Figure 6 is not truly the graph of that sequence, but it does represent the sequence in a way that might be easier to visualize.

Let's pursue that idea for other values of $c$. As Figure 7 shows, when we increase the value of $c$ the behavior of the logistic sequence becomes more complex and when $c = 4$ it is quite erratic. [This behavior is part of what it means to be *chaotic*.]

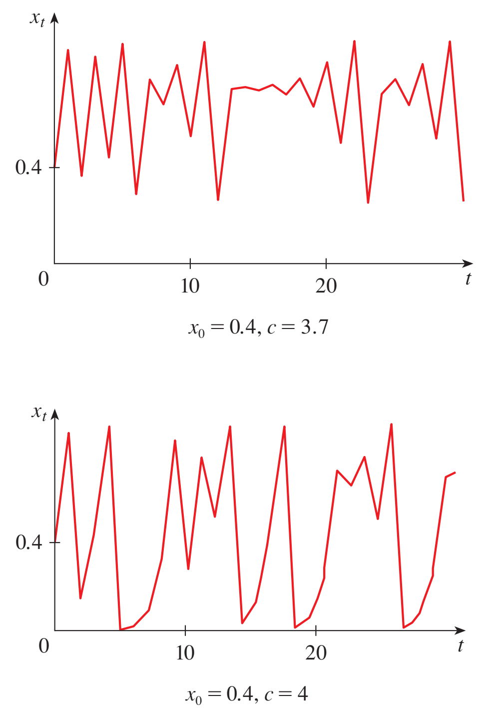


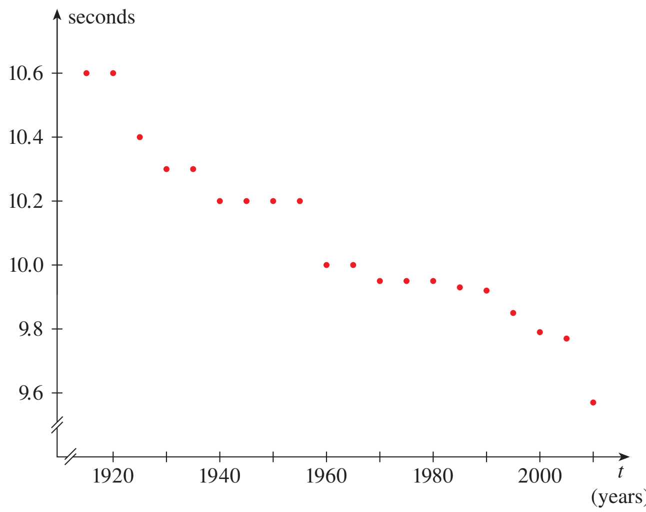

**FIGURE 7**
Logistic sequences

## EXERCISES 2.1

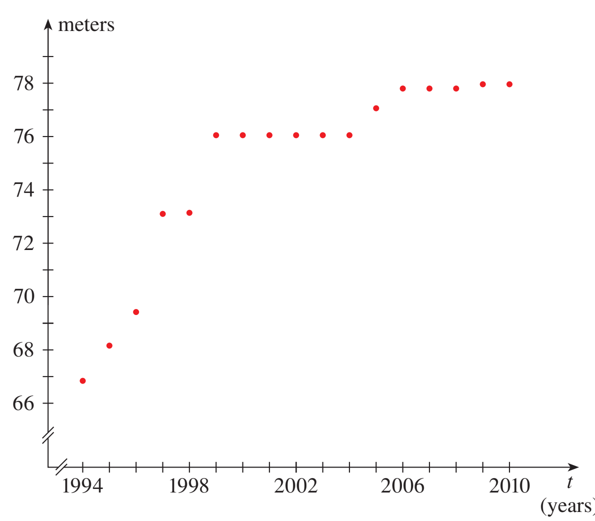
**1.** (a) What is a sequence?
(b) What does it mean to say that $\lim_{n\to\infty} a_n = 8$?
(c) What does it mean to say that $\lim_{n\to\infty} a_n = \infty$?

**2.** (a) What is a convergent sequence? Give two examples.
(b) What is a divergent sequence? Give two examples.

**3. World record sprint times** The graph plots the sequence of the world record times for the men's 100-meter sprint every five years $t$. Do you think that this sequence has a nonzero limit as $t \to \infty$? What would that mean for this sporting event?




**4. World record hammer throws** The graph plots the sequence of the world record distances for the women's hammer throw by year $t$.
(a) Explain what it would mean for this sporting event if the sequence does not have a limit as $t \to \infty$.
(b) Do you think this sequence is convergent or divergent? Explain.




**5–8** Calculate, to four decimal places, the first ten terms of the sequence and use them to plot the graph of the sequence. Does the sequence appear to have a limit? If so, calculate it. If not, explain why.

**5.** $a_n = \dfrac{n^2}{2n + 3n^2}$ **6.** $a_n = 4 - \dfrac{2}{n} + \dfrac{3}{n^2}$

**7.** $a_n = 3 + \left(-\frac{2}{3}\right)^n$ **8.** $a_n = \dfrac{n}{\sqrt{n} + 1}$

**9–26** Determine whether the sequence is convergent or divergent. If it is convergent, find the limit.

**9.** $a_n = \dfrac{1}{3n^4}$ **10.** $a_n = \dfrac{5}{3^n}$

**11.** $a_n = \dfrac{2n^2 + n - 1}{n^2}$ **12.** $a_n = \dfrac{n^3 - 1}{n}$

**13.** $a_n = \dfrac{3 + 5n}{2 + 7n}$ **14.** $a_n = \dfrac{n^3 - 1}{n^3 + 1}$

**15.** $a_n = 1 - (0.2)^n$ **16.** $a_n = 2^{-n} + 6^{-n}$

**17.** $a_n = \dfrac{n^2}{\sqrt{n^3 + 4n}}$ **18.** $a_n = \sin(n\pi/2)$

**19.** $a_n = \cos(n\pi/2)$ **20.** $a_n = \dfrac{\pi^n}{3^n}$

**21.** $a_n = \dfrac{10^n}{1 + 9^n}$ **22.** $a_n = \dfrac{\sqrt[3]{n}}{\sqrt{n} + \sqrt[4]{n}}$

**23.** $a_n = \ln(2n^2 + 1) - \ln(n^2 + 1)$

**24.** $a_n = \dfrac{3^{n+2}}{5^n}$

**25.** $a_n = \dfrac{e^n + e^{-n}}{e^{2n} - 1}$ **26.** $a_n = \ln(n + 1) - \ln n$

**27–34** Calculate, to four decimal places, the first eight terms of the recursive sequence. Does it appear to be convergent? If so, guess the value of the limit. Then assume the limit exists and determine its exact value.

**27.** $a_1 = 1,\ a_{n+1} = \frac{1}{2}a_n + 1$

**28.** $a_1 = 2,\ a_{n+1} = 1 - \frac{1}{3}a_n$

**29.** $a_1 = 2,\ a_{n+1} = 2a_n - 1$

**30.** $a_1 = 1,\ a_{n+1} = \sqrt{5a_n}$

**31.** $a_1 = 1,\ a_{n+1} = \dfrac{6}{1 + a_n}$

**32.** $a_1 = 3,\ a_{n+1} = 8 - a_n$